

PRACTICE SHEET

1. If a, b and c are real numbers, then roots of the equation $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are always:
(a) Real (b) Imaginary
(c) Positive (d) Negative
2. If $x = 2 + 2^{1/3} + 2^{2/3}$, then what is the value of $x^3 - 6x^2 + 6x$?
(a) 1 (b) 2
(c) 3 (d) -2
3. For the two equations $x^2 + mx + 1 = 0$ and $x^2 + x + m = 0$, what is/are the value/values of m for which these equations have at least one common root?
(a) Only -2 (b) Only 1
(c) -2 and 1 (d) -2 and -1
4. If $(x+a)$ is a factor of both the quadratic polynomials $x^2 + px + q$ and $x^2 + lx + m$, where p, q, l and m are constants, then which one of the following is correct?
(a) $a = \frac{(m-q)}{(l-p)}, (l \neq p)$ (b) $a = \frac{(m+q)}{(l+p)}, (l \neq p)$
(c) $l = \frac{(m-q)}{(a-p)}, (a \neq p)$ (d) $p = \frac{(m-q)}{(a-l)}, (a \neq l)$
5. If the roots of the equation $4\beta^2 + \lambda\beta - 2 = 0$ are of the form $\frac{k}{k+1}$ and $\frac{k+1}{k+2}$ then what is the value of λ ?
(a) $2k$ (b) 7
(c) 2 (d) $k+1$
6. If the sum of the squares of the roots of $x^2 - (p-2)x - (p+1) = 0$ ($p \in \mathbb{R}$) is 5, then p is equal to:
(a) 0 (b) -1
(c) 1 (d) $3/2$
7. If the roots of $x^2 - 2mx + m^2 - 1 = 0$ lie between -2 and 4, then which one of the following is correct?
(a) $-1 < m < 3$ (b) $-3 < m < 3$
(c) $3 < m < 5$ (d) $-1 < m < 5$
8. For what values of a does the equation $\cos 2x + a \sin x = 2a - 7$ possess a real solution?
(a) $a < 2$ (b) $a \geq 8$
(c) $a > 8$ (d) $2 \leq a \leq 6$
9. If $\sin \theta$ and $\cos \theta$ are the roots of $ax^2 + bx + c = 0$, then constants a, b, c will satisfy which one of the following conditions?
(a) $a^2 + b^2 + 2ac = 0$ (b) $a^2 + b^2 - 2ac = 0$
(c) $a^2 - b^2 + 2ac = 0$ (d) $-a^2 + b^2 + 2ac = 0$
10. Let $a, b \in \{1, 2, 3\}$. What is the number of equations of the form $ax^2 + bx + 1 = 0$ having real roots?
(a) 1 (b) 2
(c) 5 (d) 3
11. If $px^2 + qx + r = p(x-\alpha)(x-\beta)$, and $p^3 + pq + r = 0$; p, q and r being real numbers, then which of the following is possible:
(a) $\alpha = \beta = p$ (b) $\alpha \neq \beta = p$
(c) $\alpha = \beta \neq p$ (d) $\beta \neq \alpha = p$
12. If the equations $x^2 + k^2 = 2(k+1)x$ has equal roots, then what is the value of k ?
(a) $-\frac{1}{3}$ (b) $-\frac{1}{2}$
(c) 0 (d) 1
13. If α, β are the roots of the equations $x^2 - 2x - 1 = 0$, then what is the value of $\alpha^2\beta^{-2} + \alpha^{-2}\beta^2$?
(a) -2 (b) 0
(c) 30 (d) 34
14. Which one of the following values of x, y satisfies the equation $2x + 3y \leq 6$; $x \geq 0, y \geq 0$?
(a) $x = 0, y = 3$ (b) $x = 1, y = 2$
(c) $x = 1, y = 1$ (d) $x = 4, y = 0$
15. What is the value of x at the intersection of $y = \frac{8}{(x^2+4)}$ and $x + y = 2$?
(a) 0 (b) 1
(c) 2 (d) -1
16. If α and β are the roots of the equation $x^2 + 6x + 1 = 0$, then what is $|\alpha - \beta|$ equal to?
(a) 6 (b) $3\sqrt{2}$
(c) $4\sqrt{2}$ (d) 12
17. If $r^{1/3} + \frac{1}{r^{1/3}} = 3$ for a real number $r \neq 0$, then what $r + \frac{1}{r}$ equal to?
(a) 27 (b) 36
(c) 9 (d) 18
18. The number of rows in a lecture hall equals the number of seats in a row. If the number of rows is doubled and the number of seats in every row is reduced by 10, the number of seats is increased by 300. If x denotes the number of rows in the lecture hall, then what is the value of x ?
(a) 10 (b) 15
(c) 20 (d) 30
19. For what value of k , are the roots of the quadratic equation $(k+1)x^2 - 2(k-1)x + 1 = 0$ real and equal?
(a) $k = 0$ only (b) $k = -3$ only
(c) $k = 0$ or $k = 3$ (d) $k = 0$ or $k = -3$
20. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then what are the roots of the equation $cx^2 + bx + a = 0$?
(a) $\beta, \frac{1}{\alpha}$ (b) $\alpha, \frac{1}{\beta}$
(c) $-\alpha, -\beta$ (d) $\frac{1}{\alpha}, \frac{1}{\beta}$
21. If the roots of the equation: $(a^2 + b^2)x^2 - 2b(a+c)x + (b^2 + c^2) = 0$ are equal, then which one of the following is correct?
(a) $2b = a + c$ (b) $b^2 = ac$
(c) $b + c = 2a$ (d) $b = ac$
22. Which of the following are the two roots of the equation $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$?
(a) $1 \pm i$ (b) $2 \pm i$
(c) $1 \pm \sqrt{2}$ (d) $2 \pm i\sqrt{2}$

23. If α and β are the roots of the equation $x^2 - 2x + 4 = 0$, then what is the value of $\alpha^3 + \beta^3$?
 (a) 16 (b) -16
 (c) 8 (d) -8
24. If α and β are the roots of the equation $x^2 + x + 1 = 0$, then which of the following are the roots of the equation $x^2 - x + 1 = 0$?
 (a) α^7 and β^{13} (b) α^{13} and β^7
 (c) α^{20} and β^{20} (d) None of these
25. If the equations $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ have real roots, then what is the value of k ?
 (a) 4 (b) 8
 (c) 12 (d) 16
26. If the product of the roots of the equation $x^2 - 5x + k = 15$ is -3, then what is the value of k ?
 (a) 12 (b) 15
 (c) 16 (d) 18
27. If $\frac{1}{2 - \sqrt{-2}}$ is one of the roots of $ax^2 + bx + c = 0$, where a , b and c are real, then what are the values of a , b , c respectively?
 (a) 6, -4, 1 (b) 4, 6, -1
 (c) 3, -2, 1 (d) 6, 4, 1
28. If p and q are positive integers, then which one of the following equations has $p - \sqrt{q}$ as one of its roots?
 (a) $x^2 - 2px - (p^2 - q) = 0$
 (b) $x^2 - 2px + (p^2 - q) = 0$
 (c) $x^2 + 2px - (p^2 - q) = 0$
 (d) $x^2 + 2px + (p^2 - q) = 0$
29. If the equation $x^2 - bx + 1 = 0$ does not possess real roots, then which one of the following is correct?
 (a) $-3 < b < 3$ (b) $-2 < b < 2$
 (c) $b < 2$ (d) $b < -2$
30. If p and q are the roots of the equation $x^2 - px + q = 0$, then what are the values of p and q respectively?
 (a) 1, 0 (b) 0, 1
 (c) -2, 0 (d) -2, 1
31. If α , β are the roots of the quadratic equation $x^2 - x + 1 = 0$, then which one of the following is correct?
 (a) $(\alpha^4 - \beta^4)$ is real (b) $2(\alpha^5 + \beta^5) = (\alpha\beta)^5$
 (c) $(\alpha^6 - \beta^6) = 0$ (d) $(\alpha^8 + \beta^8) = (\alpha\beta)^8$
32. Consider the equation $(x-p)(x-6) + 1 = 0$ having integral coefficients. If the equation has integral roots, then what values can p have?
 (a) 4 or 8 (b) 5 or 10
 (c) 6 or 12 (d) 3 or 6
33. The solution set of inequality $\frac{x-4}{x-3} \leq 2$, is
 (a) $(-\infty, 3) \cup (10, \infty)$ (b) $(3, 10]$
 (c) $(-\infty, 3) \cup [10, \infty)$ (d) $(-\infty, 3] \cup [10, \infty)$
34. The solution set of inequality $\frac{|x-2|}{x-2} < 0$, is
 (a) $(2, \infty)$ (b) $(-2, \infty)$
 (c) $\mathbb{R}$ (d) $(-2, 2)$
35. If $|x-1| + |x| + |x+1| \geq 6$, then x belongs to
 (a) $(-\infty, 3]$ (b) $(-\infty, -2] \cup [2, \infty)$
 (c) $\mathbb{R}$ (d) Null set
36. If $2 - 3x - 2x^2 \geq 0$, then
 (a) $x \leq -2$ (b) $-2 \leq x \leq 1/2$
 (c) $x \geq -2$ (d) $x \leq 1/2$
37. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then what is the value of expression $(\alpha + 1)(\beta + 1)$?
 (a) $\frac{a+b+c}{a}$ (b) $\frac{b+c-a}{a}$
 (c) $\frac{a-b+c}{a}$ (d) $\frac{a+b-c}{a}$
38. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b}$ is:
 (a) $\frac{a}{bc}$ (b) $\frac{b}{ac}$
 (c) $\frac{c}{ab}$ (d) $\frac{1}{abc}$
39. If the sum of the roots of the equation $x^2 - k^2x + 30kx - 161x - 64 = 0$ is zero, then what is the difference of the roots?
 (a) 15 (b) 16
 (c) 17 (d) 183
40. What is the greatest value of k for which $2x^2 - 4x + k = 0$ has real roots?
 (a) 1 (b) 2
 (c) 3 (d) 4

ANSWER KEY

1.	a	2.	b	3.	c	4.	a	5.	b	6.	c	7.	b	8.	b	9.	c	10.	d
11.	a	12.	b	13.	d	14.	c	15.	a	16.	c	17.	d	18.	d	19.	c	20.	d
21.	b	22.	b	23.	b	24.	d	25.	d	26.	a	27.	a	28.	b	29.	b	30.	a
31.	c	32.	a	33.	c	34.	b	35.	b	36.	b	37.	c	38.	b	39.	b	40.	b

Solutions

Sol.1. (a)

Given,

$$(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$$

$$\Rightarrow 3x^2 - 2(b+a+c)x + ab + bc + ca = 0$$

Now, D =

$$\sqrt{4(a+b+c)^2 - 12(ab+bc+ca)}$$

$$\Rightarrow 2\sqrt{a^2+b^2+c^2-ab-bc-ca}$$

$$= 2\sqrt{\frac{1}{2}\{(a-b)^2 + (b-c)^2 + (c-a)^2\}} \geq 0$$

= 2 i.e. positive, so roots are real.

Sol.2. (b)

$$x = 2 + 2^{1/3} + 2^{2/3}$$

$$(x-2) = 2^{1/3} + 2^{2/3}$$

Taking whole cube both side

$$(x-2)^3 = (2^{1/3} + 2^{2/3})^3$$

$$x^3 - 8 - 6x^2 + 12x = 2 + 4 + 6(2^{1/3} + 2^{2/3})$$

$$x^3 - 8 - 6x^2 + 12x = 2 + 4 + 6(x-2)$$

$$x^3 - 6x^2 + 6x = 2$$

Sol.3. (c)

Suppose the given equations have a common root α .

$$\text{Then, } \alpha^2 + m\alpha + 1 = 0 \quad \dots\dots(i)$$

$$\text{and } \alpha^2 + \alpha + m = 0 \quad \dots\dots(ii)$$

subtract (ii) from (i)

$$m\alpha - \alpha + 1 - m = 0$$

$$\alpha(m-1) = m-1$$

$$\alpha = 1$$

By first and last

$$\frac{\alpha}{m^2-1} = \frac{1}{1-m}$$

$$\Rightarrow 1-m = m^2-1 \quad (\because \alpha = 1)$$

$$\Rightarrow m^2 + m - 2 = 1$$

$$\Rightarrow m^2 + m - 2 = 0$$

$$\Rightarrow (m+2)(m-1) = 0$$

$$\therefore m = 1, -2$$

Sol.4. (a)

Since, $(x+a)$ is a factor of $x^2 + px + q$ and $x^2 + lx + m$. so we can say that $x = -a$ is a root of both equations

$$\therefore a^2 - ap + q = 0 \quad \dots(i)$$

$$\text{and } a^2 - la + m = 0 \quad \dots(ii)$$

subtract Eq. (ii) from (i), we get

$$-ap + q + la - m = 0 \Rightarrow (l-p)a = m-q$$

$$a = \frac{m-q}{l-p}, \quad (l \neq p)$$

Sol.5. (b)

Let $\frac{k}{k+1}$ and $\frac{k+1}{k+2}$ are the roots of the equation

$$4\beta^2 + \lambda\beta - 2 = 0, \text{ then}$$

Sum of the roots

$$\Rightarrow \frac{k}{k+1} + \frac{k+1}{k+2} = -\frac{\lambda}{4}$$

and product of the roots, $\frac{k}{k+1} \times \frac{k+1}{k+2} = -\frac{2}{4}$

$$\Rightarrow \frac{k}{k+2} = -\frac{1}{2} \Rightarrow 2k = -k-2 \Rightarrow k = -\frac{2}{3}$$

Putting the value of k in (i), we get

$$\frac{-\frac{2}{3}}{\frac{-2}{3}+1} + \frac{-\frac{2}{3}+1}{\frac{-2}{3}+2} = \frac{\lambda}{4}$$

$$\Rightarrow \frac{-\frac{2}{3}}{\frac{1}{3}} + \frac{\frac{1}{3}}{\frac{4}{3}} = -\frac{\lambda}{4} \Rightarrow -2 + \frac{1}{4} = -\frac{\lambda}{4}$$

$$\Rightarrow \lambda = 7$$

Sol.6. (c)

Let α and β be the roots of

$$x^2 - (p-2)x - (p+1) = 0,$$

Then $\alpha + \beta = (p-2)$ and $\alpha\beta = -(p+1)$

$$\therefore \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$\Rightarrow (\alpha + \beta)^2 - 2\alpha\beta = 5$$

$$\Rightarrow (p-2)^2 + 2(p+1) = 5$$

$$\Rightarrow p^2 - 4p + 4 + 2p + 2 = 5$$

$$\Rightarrow p^2 - 2p + 1 = 0$$

$$\Rightarrow (p-1)^2 = 0 \Rightarrow p = 1$$

Sol.7. (b)

Since, the roots of $x^2 - 2mx + m^2 - 1 = 0$ lie between -2 and 4

i.e. roots are real

$$b^2 - 4ac \geq 0$$

$$(2m)^2 - 4(m^2 - 1) \geq 0$$

$$\Rightarrow 4m^2 - 4m^2 + 4 \geq 0 \quad \text{where } m \in \mathbb{R}$$

$$\Rightarrow 4 \geq 0, \text{ roots are always real.}$$

If both roots lie between -2 and 4

$$\text{Then } f(-2)f(4) > 0$$

$$(4 + 4m + m^2 - 1)(16 - 8m + m^2 - 1) > 0$$

$$(m+1)(m+3)(m-3)(m-5) > 0$$

By number line

$$m \in (-\infty, -5) \cup (-3, 3) \cup (5, \infty)$$

$$\text{so } -3 < m < 3$$

Sol.8. (d)

Given equations $\cos 2x + a \sin x = 2a - 7$ can be written as

$$\cos^2 x - \sin^2 x + a \sin x = 2a - 7$$

$$\Rightarrow 1 - \sin^2 x - \sin^2 x + a \sin x = 2a - 7$$

$$\Rightarrow 2 \sin^2 x - a \sin x + (2a - 7 - 1) = 0$$

$$\Rightarrow 2 \sin^2 x - a \sin x + 2a - 8 = 0$$

This is a quadratic equation in $\sin x$ and its discriminant ≥ 0

$$\text{Here, } a = 2, b = -a, c = 2a - 8$$

$$\Rightarrow a^2 - 4.2.(2a - 8) \geq 0$$

$$\Rightarrow a^2 - 16a + 64 \geq 0$$

$$\Rightarrow (a - 8)^2 \geq 0, \text{ that is always correct.}$$

Roots of Q.E. are

$$-1 \leq \frac{a \pm \sqrt{(a-8)^2}}{4} \leq 1$$

$$-1 \leq \frac{a \pm (a-8)}{4} \leq 1$$

$$-4 \leq a \pm (a-8) \leq 4$$

For positive sign

$$2 \leq a \leq 6$$

Sol.9. (c)

As given $\sin \theta$ and $\cos \theta$ are the roots of the equation $ax^2 + bx + c = 0$

So, sum of roots

$$= \sin \theta + \cos \theta = -\frac{b}{a} \quad \dots(1)$$

and product of roots

$$= \sin \theta \cos \theta = \frac{c}{a} \quad \dots(2)$$

On squaring both sides in Eq. (1), we get

$$(\sin \theta + \cos \theta)^2 = \frac{b^2}{a^2}$$

$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{b^2}{a^2}$$

$$\Rightarrow 1 + \frac{2c}{a} = \frac{b^2}{a^2}$$

$$\Rightarrow a^2 + 2ca = b^2 \quad (\text{using Equations 2})$$

$$\Rightarrow a^2 - b^2 + 2ac = 0$$

Sol.10. (d)

The given equations is $ax^2 + bx + 1 = 0$

This equation has real roots

When discriminant ≥ 0

$$\therefore b^2 - 4a \geq 0$$

$$\Rightarrow b^2 \geq 4a$$

a, b has to be selected from, three numbers, so total 3 selections are possible when (a, b) are $(1, 2), (1, 3)$ and $(2, 3)$.

Thus, the number of equations of the form $ax^2 + bx + 1 = 0$ having real root is 3.

Sol.11. (a)

Given equations is

$$px^2 + qx + r = p(x - \alpha)(x - \beta)$$

$$\Rightarrow px^2 + qx + r = px^2 - p(\alpha + \beta)x + \alpha\beta p$$

Compare coeff. Of x and constant term.

$$\Rightarrow \alpha\beta p = r \text{ and } q = -(\alpha + \beta)p$$

Also given that

$$p^3 + pq + r = 0$$

Putting value of q and r from (1)

$$\Rightarrow p^3 - p^2(\alpha + \beta) + \alpha\beta p = 0$$

$$\Rightarrow p^2 - p(\alpha + \beta) + \alpha\beta = 0$$

$$\Rightarrow (p - \alpha)(p - \beta) = 0 \Rightarrow \alpha = \beta = p$$

Sol.12. (b)

$$x^2 + k^2 = 2(k+1)x$$

$$\Rightarrow x^2 - 2(k+1)x + k^2 = 0$$

For roots to be equal discriminate = 0

$$\text{So, } \{-2(k+1)\}^2 - 4k^2 = 0$$

$$\text{or, } 4(k+1)^2 - 4k^2 = 0$$

$$\text{or, } (k+1)^2 - k^2 = 0$$

$$\text{or, } 2k+1=0 \Rightarrow k=-\frac{1}{2}$$

Sol.13. (d)

Since, α and β are the roots of the equation $x^2 - 2x - 1 = 0$, then

Sum of roots, $\alpha + \beta = 2$ and

Product of the roots $\alpha\beta = -1$

$$\text{Since, } (\alpha + \beta) = \alpha^2 + \beta^2 + 2\alpha\beta$$

$$\Rightarrow 4 = \alpha^2 + \beta^2 - 2$$

$$\Rightarrow \alpha^2 + \beta^2 = 6$$

$$\text{Now, } \alpha^2\beta^2 + \alpha^2\beta^2 = \frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2} = \frac{\alpha^4 + \beta^4}{(\alpha\beta)^2}$$

$$\Rightarrow (\alpha^2 + \beta^2)^2 = 6^2$$

$$\Rightarrow \alpha^4 + \beta^4 + 2\alpha^2\beta^2 = 36$$

$$\Rightarrow \alpha^4 + \beta^4 + 2 = 36 \quad (\because \alpha\beta = -1)$$

$$\Rightarrow \alpha^4 + \beta^4 = 34 \quad \dots(i)$$

$$\Rightarrow \frac{\alpha^4 + \beta^4}{(\alpha\beta)^2} = \frac{34}{(-1)^2} = 34$$

Putting value of $\alpha^4 + \beta^4 = 34$ from Equation (i)

Sol.14. (c)

There can be many values of x and y for this in equation. In the given options only $x=1, y=1$ satisfy the given equation.

Sol.15. (a)

Given equations are

$$Y = \frac{8}{x^2 + 4} \text{ and } x + y = 2$$

Putting value of y from 1st equation into second equations

$$x + \frac{8}{x^2 + 4} = 2$$

$$\Rightarrow x^3 + 4x + 8 = 2x^2 + 8$$

$$\Rightarrow x^3 - 2x^2 + 4x = 0$$

$$\Rightarrow x(x^2 - 2x + 4) = 0$$

$$\Rightarrow x = 0$$

[The other value of x is not real]

Sol.16. (c)

α and β are the roots of the equation $x^2 + 6x + 1 = 0$

$$\Rightarrow \alpha + \beta = -6 \text{ and } \alpha\beta = 1$$

$$\text{Now, } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= (-6)^2 - 4 = 36 - 4 = 32$$

$$\Rightarrow |\alpha - \beta| = \sqrt{32} = 4\sqrt{2}$$

Sol.17. (d)

Given equation is:

$$r^{1/3} + \frac{1}{r^{1/3}} = 3$$

Cubing both sides, we get

$$\left(r^{1/3} + \frac{1}{r^{1/3}}\right)^3 = 3^3$$

$$[(a+b)^3 = a^3 + b^3 + 3ab(a+b)]$$

$$\Rightarrow r + \frac{1}{r} + 3\left(r^{1/3} + \frac{1}{r^{1/3}}\right)^3 = 27$$

$$\Rightarrow r + \frac{1}{r} + 3.3 = 27 \Rightarrow r + \frac{1}{r} + 27 - 9 = 18$$

Sol.18. (d)

As given

$$\Rightarrow \text{Number of rows} = x$$

$$\Rightarrow \text{Number of seats in each row} = x$$

$$\text{Total number of seats in the hall} = x^2$$

$$\text{Revised number of rows} = 2x$$

Revised number of rows = $2x$

Revised number in each row = $x - 10$

$$\text{Thus Revised number of seats} = 2x(x-10) = 2x^2 - 20x$$

According to questions

$$2x^2 - 20x = 300 + x^2$$

$$\Rightarrow x^2 - 20x - 300 = 0$$

$$\Rightarrow x^2 - 30x + 10x - 300 = 0$$

$$\Rightarrow (x-30)(x+10) = 0$$

$$\Rightarrow x = 30 (\because x \neq 0)$$

Sol.19. (c)

As given:

Roots of the quadratic equation

$$(k+1)x^2 - 2(k-1)x + 1 = 0 \text{ are real and equal,}$$

Its discriminate

$$\{-2(k-1)\}^2 - 4(k+1) = 0$$

$$\Rightarrow 4(k^2 - 2k + 1) - 4(k+1) = 0$$

$$\Rightarrow k^2 - 2k + 1 - k - 1 = 0$$

$$\Rightarrow k^2 - 3k = 0$$

$$\Rightarrow k = 0, k = 3$$

Sol.20. (d)

As given, α , and β are the roots of $ax^2 + bx + c = 0$, then the roots of $cx^2 + bx + a = 0$ will be reciprocal of α and β , i.e.,

$$\frac{1}{\alpha} \text{ and } \frac{1}{\beta}$$

Sol.21. (b)

Since, roots of the given equation are equal.

$$\therefore D = 0$$

On solving, we get $b^2 = ac$

Sol.22. (b)

$$(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$$

$$\Rightarrow \left(\frac{x^2 + 2}{x}\right)^2 - 6\left(\frac{x^2 + 2}{x}\right) + 8 = 0$$

$$\Rightarrow \left(\frac{x^2 + 2}{x} - 2\right)\left(\frac{x^2 + 2}{x} - 4\right) = 0$$

$$\Rightarrow \frac{x^2 + 2}{x} = 2 \Rightarrow x^2 + 2 = 2x$$

$$\Rightarrow x^2 - 2x + 2 = 0 \Rightarrow D \leq 0$$

$$\Rightarrow \frac{x^2 + 2}{x} = 4 \Rightarrow x^2 + 2 = 4x$$

$$\Rightarrow x^2 - 4x + 2 = 0 \Rightarrow D \geq 0$$

$$\Rightarrow x^2 - 2x + 2 = 0$$

$$\Rightarrow 2 \pm i$$

Sol.23. (b)

$$\text{Use } \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$$

Sol.24. (d)

Since, α and β be the roots of the equation $x^2 + x + 1 = 0$

$$\therefore \alpha + \beta = -1 \text{ and } \alpha\beta = 1$$

$$\Rightarrow \alpha = \omega \text{ and } \beta = \omega^2$$

Since none of the options (a), (b), (c) satisfy the given equation, hence option (d) is correct.

Sol.25. (d)

Since, equations $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ have real roots

$$\therefore k^2 \geq 4 \times 64 \Rightarrow k \geq 16 \quad \dots(i)$$

$$\text{and } 64 \geq 4k \Rightarrow k \leq 16 \quad \dots(ii)$$

From eqs. (i) and (ii),

$$\therefore k = 16$$

Sol.26. (a)

$$k - 15 = -3 \Rightarrow k = 12$$

Sol.27. (a)

$$\text{he given root is } = \frac{1}{2 - \sqrt{-2}} = \frac{2 + \sqrt{2}i}{6}$$

$$\therefore \text{Another root} = \frac{2 - \sqrt{2}i}{6}$$

$$\therefore \text{equation is } -x^2 - \frac{2}{3}x + \frac{1}{6} = 0 \text{ or } 6x^2 - 4x + 1 = 0$$

0

Thus, $a = 6, b = -4$ and $c = 1$

Sol.28. (b)

sum of roots = $2p$

product of roots = $p^2 - q$

Sol.29. (b)

Since, given equation does not possess real root.

$$\therefore D < 0 \Rightarrow b^2 - 4 < 0$$

$$\Rightarrow b^2 < 4 \Rightarrow -2 < b < 2$$

Sol.30. (a)

$$x^2 - px + q = 0$$

p is root of this equation

$$p^2 - p^2 + q = 0 \Rightarrow q = 0$$

$$\text{and } pq = q \Rightarrow p = 1$$

Sol.31. (c)

Since, α and β be the roots of the equation $x^2 - x + 1 = 0$

$$\therefore \alpha + \beta = 1 \text{ and } \alpha\beta = 1$$

$$\text{Now, } \alpha - \beta = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{3}i$$

$$\Rightarrow \alpha = \frac{1 + i\sqrt{3}}{2} \text{ and } \beta = \frac{1 - i\sqrt{3}}{2}$$

$$\Rightarrow \alpha = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} - \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$$

$$(a) \alpha^4 - \beta^4 = \cos$$

$$\frac{4\pi}{3} + i \sin \frac{4\pi}{3} - \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$$

$$= 2i \sin \frac{4\pi}{3}$$

$$\Rightarrow \alpha^4 - \beta^4 \text{ is not real}$$

(b) 2(\alpha^5 + \beta^5)

$$= 2 \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} + \cos \frac{5\pi}{3} - i \sin \frac{5\pi}{3} \right)$$

$$= 2.2 \cos \frac{5\pi}{3} = 4 \cdot \frac{1}{2} = 2$$

Now, $(\alpha\beta)^5 = 1$

$$\Rightarrow 2(\alpha^5 + \beta^5) \neq (\alpha\beta)^5$$

(c) $\alpha^6 - \beta^6 = \cos$

$$\frac{6\pi}{3} + i \sin \frac{6\pi}{3} - \cos \frac{6\pi}{3} + i \sin \frac{6\pi}{3}$$

$$= 2 \text{ is in } 2\pi = 0$$

Hence, option (c) is correct.

Sol.32. (a)

The given equation can be rewritten as

$$x^2 - (p+6)x + (6p+1) = 0$$

Now, $b^2 - 4ac = 0$ ($\because$ equation has integral roots)

$$\Rightarrow (p+6)^2 - 4(6p+1) = 0$$

$$\text{Now } p^2 - 12p + 32 = 0$$

$$\Rightarrow (p-4)(p-8) = 0$$

$$\therefore p = 4, 8$$

Sol.33. (c)

$$\frac{x+4}{x-3} \leq 2$$

$$\frac{x+4}{x-3} - 2 \leq 0$$

$$\frac{x+4-2x+6}{x-3} \leq 0$$

$$\frac{x-10}{x-3} \geq 0$$

Now by number line

$$x \in (-\infty, 3) \cup [10, \infty)$$

Sol.34. (d)

$$\frac{|x-2|}{x-2} < 0$$

Numerator is always positive so
Denominator $x-2 < 0$

$$x < 2$$

Sol.35. (b)

$$|x-1| + |x| + |x+1| \geq 6$$

If $x > 1$

$$x-1+x+x+1 \geq 6$$

$$3x \geq 6$$

$$x \geq 2$$

....(i)

If $x < -1$

$$-x+1-x-x-1 \geq 6$$

$$-3x \geq 6$$

$$x \leq -2$$

....(ii)

by (i) and (ii)

$$x \in (-\infty, -2] \cup [2, \infty)$$

Sol.36. (b)

$$2-3x-2x^2 \geq 0$$

$$2x^2+3x-2 \leq 0$$

$$2x^2+4x-x-2 \leq 0$$

$$2x(x+2)-(x+2) \leq 0$$

$$(x+2)(2x-1) \leq 0$$

Now by number line

$$-2 \leq x \leq 1/2$$

Sol.37. (c)

$$(\alpha+1)(\beta+1)$$

$$\alpha\beta + \alpha + \beta + 1$$

$$\frac{c}{a} - \frac{b}{a} + 1 = \frac{a-b+c}{a}$$

Sol.38. (b)

α and β are the roots of the equation $ax^2+bx+c=0$

$$\text{so } a\alpha^2+b\alpha+c=0 \Rightarrow \alpha(a\alpha+b)+c=0$$

$$(a\alpha+b)=-c/\alpha \quad \dots\dots(i)$$

$$\text{and } ax^2+bx+c=0 \Rightarrow \beta(a\beta+b)+c=0$$

$$(a\beta+b)=-c/\beta \quad \dots\dots(ii)$$

Multiply (i) and (ii)

$$\frac{1}{a\alpha+b} + \frac{1}{a\beta+b} = \left(-\frac{\alpha}{c}\right) + \left(-\frac{\beta}{c}\right)$$

$$= -\left(\frac{\alpha+\beta}{c}\right) = \frac{b}{ac}$$

Sol.39. (b)

sum of the roots is zero, i.e. coefficient of x is zero
then equation will be $x^2-64=0$

so $x=8$ and -8

difference of roots = 16

Sol.40. (b)

$2x^2-4x+k=0$ has real roots

$$b^2-4ac \geq 0$$

$$16-8k \geq 0$$

$$2 \geq k$$

Maximum value of k is 2

NDA PYQ

1. One of the root of the quadratic equations $ax^2 + bx + c = 0$, $a \neq 0$ is positive and the other is negative. The condition for this to happen is:
 (a) $a > 0, b > 0, c > 0$ (b) $a > 0, b < 0, c > 0$
 (c) $a < 0, b > 0, c < 0$ (d) $a < 0, c > 0$
[NDA (I) 2011]
2. What is the sum of the roots of the equation $(2 - \sqrt{3})x^2 - (7 - 4\sqrt{3})x + (2 + \sqrt{3}) = 0$?
 (a) $2 - \sqrt{3}$ (b) $2 + \sqrt{3}$
 (c) $7 - 4\sqrt{3}$ (d) 4
[NDA (I) 2011]
3. If one root of the equation $ax^2 + bx + c = 0$, $a \neq 0$ is reciprocal of the other root, then which one of the following is correct?
 (a) $a = c$ (b) $b = c$
 (c) $a = -c$ (d) $b = 0$
[NDA (I) 2011]
4. If sum of squares of the roots of the equation $x^2 + kx - b = 0$ is $2b$, then what is k equal to?
 (a) 1 (b) b
 (c) $-b$ (d) 0
[NDA (I) 2011]
5. If 3 is the root of the equation $x^2 - 8x + k = 0$, then what is the value of k ?
 (a) -15 (b) 9
 (c) 15 (d) 24
[NDA (I) 2011]
6. What is the condition that one root of the equation $ax^2 + bx + c = 0$, $a \neq 0$ should be double the other?
 (a) $2a^2 = 9bc$ (b) $2b^2 = 9ac$
 (c) $2c^2 = 9ab$ (d) None of these
[NDA (I) 2011]
7. What is the solution set for the equation $x^4 - 26x^2 + 25 = 0$?
 (a) $\{-5, -1, 1, 5\}$ (b) $\{-5, -1\}$
 (c) $\{1, 5\}$ (d) $\{-5, 0, 1, 5\}$
[NDA (I) 2011]
8. If p, q, r are rational numbers, then the roots of the equation $x^2 - 2px + p^2 - q^2 + 2qr - r^2 = 0$ are:
 (a) Complex (b) Pure imaginary
 (c) Irrational (d) Rational
[NDA (I) 2011]
9. If $x + y \leq 4$, then the how many non-zero positive integer ordered pair (x, y) ?
 (a) 4 (b) 5
 (c) 6 (d) 8
[NDA (I) 2011]
10. If α and β are the roots of equation $4x^2 + 3x + 7 = 0$, then what is the value of $\alpha^{-2} + \beta^{-2}$?
 (a) $47/49$ (b) $49/47$
 (c) $-47/49$ (d) $-49/47$
[NDA (I) 2011]
11. What is the value of $y = \sqrt{8 + 2\sqrt{8 + 2\sqrt{8 + 2\sqrt{8 + \dots}}}}$
 (a) 10 (b) 8
 (c) 6 (d) 4
[NDA (II) 2011]
12. If α and β be the roots of the equation $(x - a)(x - b) = c$, $c \neq 0$. Then, the roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are:
 (a) a, c (b) b, c
 (c) a, b (d) $a + b, a + c$
[NDA (II) 2011]
13. The equation $x^2 - 4x + 29 = 0$ has one root $2 + 5i$. What is the other root? ($i = \sqrt{-1}$)
 (a) 2 (b) 5
 (c) $2 + 5i$ (d) $2 - 5i$
[NDA (II) 2011]
14. One of the roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ is 1, then what is the second root?
 (a) $\frac{b(c-a)}{a(b-c)}$ (b) $\frac{b(c-a)}{a(b-c)}$
 (c) $\frac{c(a-b)}{a(b-c)}$ (d) $-\frac{c(a-b)}{a(b-c)}$
[NDA (II) 2011]
15. What are the roots of the equation:
 $2(y + 2)^2 - 5(y + 2) = 12$?
 (a) $-7/2, 2$ (b) $-3/2, 4$
 (c) $-5/3, 3$ (d) $3/2, 4$
[NDA (II) 2011]
16. If the roots of the equation $3x^2 - 5x + q = 0$ are equal, then what is the value of q ?
 (a) 2 (b) $5/12$
 (c) $12/25$ (d) $25/12$
[NDA (II) 2011]
17. If the equations $x^2 - px + q = 0$ and $x^2 - ax + b = 0$ have a common root and the roots of the second equation are equal, then which one of the following is correct?
 (a) $aq = 2(b + p)$ (b) $aq = b + p$
 (c) $ap = 2(b + q)$ (d) $ap = b + q$
[NDA (II) 2011]
18. If the roots of equation $x^2 - 4x - \log_3 N = 0$, are real, then what is the minimum value of N ?
 (a) $1/256$ (b) $1/27$
 (c) $1/64$ (d) $1/81$
[NDA (II) 2011]
19. The equation $\tan^4 x - 2\sec^2 x + a^2 = 0$ will have at-least one real solution if
 (a) $|a| \leq 4$ (b) $|a| \leq 4$
 (c) $|a| \leq \sqrt{3}$ (d) none of above
[NDA (II) 2011]
20. If $\sin \theta = x + \frac{a}{x}$ for all $x \in \mathbb{R} - \{0\}$, then which one of the following is correct?
 (a) $a \geq 4$ (b) $a \geq \frac{1}{2}$
 (c) $a \leq \frac{1}{4}$ (d) $a \leq \frac{1}{2}$
[NDA (II) 2011]
21. What is the sum of the squares of the roots of the equation $x^2 + 2x - 143 = 0$?
 (a) 170 (b) 180
 (c) 190 (d) 290
[NDA (I) 2012]
22. If one of the roots of the equation $x^2 + ax - b = 0$ is, 1, then what is the value of $(a-b)$?

- (a) -1 (b) 1
(c) 2 (d) -2
[NDA (I) 2012]
23. If α and β are the roots of the equation. $x^2 - q(1+x) - r = 0$ then what is the value of $(1+\alpha)(1+\beta)$?
(a) $1-r$ (b) $q-r$
(c) $1+r$ (d) $q+r$
[NDA (I) 2012]
24. The solution of the simultaneous linear equations $2x+y=6$ and $3y=8+4x$ will also be satisfied by which one of the following linear equation?
(a) $x+y=5$ (b) $2x+y=5$
(c) $2x-3y=10$ (d) $2x+3y=6$
[NDA (I) 2012]
25. If the difference between the roots of $ax^2+bx+c=0$ is 1, then which one of the following is correct?
(a) $b^2 = a(a+4c)$ (b) $a^2 = b(b+4c)$
(c) $a^2 = c(a+4c)$ (d) $b^2 = a(b+4c)$
[NDA (I) 2012]
- Direction (for next two):** The equation formed by multiplying each root of $ax^2+bx+c=0$ by 2 is $x^2+36x+24=0$.
26. What is the value of $b:c$?
(a) 3:1 (b) 1:2
(c) 1:3 (d) 3:2
[NDA (I) 2012]
27. Which one of the following correct?
(a) $bc = a^2$ (b) $bc = 36a^2$
(c) $bc = 72a^2$ (d) $bc = 108a^2$
[NDA (I) 2012]
28. If the roots of the quadratic equation $3x^2 - 5x + p = 0$ are real and unequal, then which one of the following is correct?
(a) $p = 25/12$ (b) $p < 25/12$
(c) $p > 25/12$ (d) $p \leq 25/12$
[NDA (II) 2012]
29. If the roots of a quadratic equation are $m+n$ and $m-n$, then the quadratic equation will be:
(a) $x^2 + 2mx + m^2 - mn + n^2 = 0$
(b) $x^2 + 2mx + (m-n)^2 = 0$
(c) $x^2 - 2mx + m^2 - n^2 = 0$
(d) $x^2 + 2mx + m^2 - n^2 = 0$
[NDA (II) 2012]
30. If α, β are the roots of $x^2 + px - q = 0$ and γ, δ are the roots of $x^2 - px + r = 0$, then what is the value of $(\beta + \gamma)(\beta + \delta)$?
(a) $p+r$ (b) $p+q$
(c) $q+r$ (d) $p-q$
[NDA (II) 2012]
31. If α and β are roots of the equation $x^2 + bx + c = 0$, then what is the value of $\alpha^{-1} + \beta^{-1}$?
(a) $-\frac{b}{c}$ (b) $\frac{b}{c}$
(c) $\frac{c}{b}$ (d) $-\frac{c}{b}$
[NDA (I) 2013]
32. $(x+1)^2 - 1 = 0$ has
(a) One real root (b) Two real roots
(c) Two imaginary roots (d) Four real roots
[NDA (I) 2013]
33. If $4^x - 6 \cdot 2^x + 8 = 0$, then the values of x are:
(a) 1, 2 (b) 1, 1
(c) 1, 0 (d) 2, 2
[NDA (I) 2013]
34. If the roots of a quadratic equation $ax^2 + bx + c = 0$ are α and β , then the quadratic equation having roots α^2 and β^2 is:
(a) $x^2 - (b^2 - 2ac)x + c = 0$
(b) $a^2 x^2 - (b^2 - 2ac)x + c = 0$
(c) $ax^2 - (b^2 - 2ac)x + c^2 = 0$
(d) $a^2 x^2 - (b^2 - 2ac)x + c^2 = 0$
[NDA (I) 2013]
35. If the sum of the roots of a quadratic equation is 3 and the product is 2, then the equation is:
(a) $2x^2 - x + 3 = 0$ (b) $x^2 - 3x + 2 = 0$
(c) $x^2 + 3x + 2 = 0$ (d) $x^2 + 3x - 2 = 0$
[NDA (I) 2013]
36. What is the degree of the equation $\frac{1}{x-3} = \frac{1}{x+2} - \frac{1}{2}$?
(a) 0 (b) 1
(c) 2 (d) 3
[NDA (I) 2013]
37. If the roots of the equation $3ax^2 + 2bx + c = 0$ are in the ratio 2:3, then which one of the following is correct?
(a) $8ac = 25b^2$ (b) $8ac = 9b^2$
(c) $8b^2 = 9ac$ (d) $8b^2 = 25ac$
[NDA (I) 2013]
38. The area of a rectangle whose length is five more than twice of its width is 75 square unit. The length is
(a) 5 unit (b) 10 unit
(c) 15 unit (d) 20 unit
[NDA (I) 2013]
39. The quadratic equation $x^2 + bx + 4 = 0$ will have real roots, if:
(a) Only $b \leq -4$ (b) Only $b \geq 4$
(c) $-4 < b < 4$ (d) $b \leq -4, b \geq 4$
[NDA (II) 2013]
40. If α and β are the roots of the equation $x^2 + x + 2 = 0$, then what is $\frac{\alpha^{10} + \beta^{10}}{\alpha^{-10} + \beta^{-10}}$ equal to?
(a) 4096 (b) 2048
(c) 1024 (d) 512
[NDA (II) 2013]
41. What is the difference in the roots of the equation $x^2 - 10x + 9 = 0$.
(a) 2 (b) 3
(c) 5 (d) 8
[NDA (II) 2013]
42. If α and β are the roots of the equation $ax^2 + bx + b = 0$, then what is the value of $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} = ?$
(a) -10 (b) 0
(c) 1 (d) 2
[NDA (II) 2013]
43. The roots of the equation $x^2 - 8x + 16 = 0$:
(a) Are imaginary (b) Are distinct and real
(c) Are equal and real (d) Cannot be determined
[NDA (II) 2013]
44. If a and b are rational and b is not perfect square, then the quadratic equation with rational coefficients whose one root is $3a + \sqrt{b}$ is:
(a) $x^2 - 6ax + 9a^2 - b = 0$ (b) $3ax^2 + x - \sqrt{b} = 0$
(c) $x^2 + 3ax + \sqrt{b} = 0$ (d) $\sqrt{b} x^2 + x - 3a = 0$
[NDA (II) 2013]

45. How many real roots does the quadratic equation $f(x) = x^2 + 3|x| + 2 = 0$ have?
 (a) One (b) Two
 (c) Four (d) No real root
[NDA (II) 2013]
46. If $8x - 9y = 20$, and $7x - 10y = 9$, then what is $2x - y$ equal to?
 (a) 10 (b) 11
 (c) 12 (d) 13
[NDA (II) 2013]
47. What is the positive square root of $7 + 4\sqrt{3}$?
 (a) $\sqrt{3} - 1$ (b) $\sqrt{3} + 1$
 (c) $\sqrt{3} - 2$ (d) $\sqrt{3} + 2$
[NDA-2013(2)]
48. If α and β are the roots the equation $ax^2 + bx + c = 0$, where $a \neq 0$, then $(a\alpha + b)(a\beta + b)$ is equal to:
 (a) ab (b) bc
 (c) ca (d) abc
[NDA (I) 2014]
49. The roots of the equation $2a^2x^2 - 2abx + b^2 = 0$, when $a < 0$ and $b > 0$ are:
 (a) Sometimes complex (b) Always irrational
 (c) Always complex (d) Always real
[NDA (I) 2014]
50. Every quadratic equation $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}$, $a \neq 0$ has:
 (a) Exactly one real root (b) At least one real root
 (c) At least two real roots (d) At most two real roots
[NDA (II) 2014]
51. If α, β are roots of $ax^2 + bx + c = 0$ and $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0$, then what is h equal to?
 (a) $\frac{1}{2} \left(\frac{b}{a} - \frac{q}{p} \right)$ (b) $\frac{1}{2} \left(-\frac{b}{a} + \frac{q}{p} \right)$
 (c) $\frac{1}{2} \left(\frac{b}{p} + \frac{q}{a} \right)$ (d) $\frac{1}{2} \left(-\frac{b}{a} + \frac{q}{a} \right)$
[NDA (II) 2014]
52. If $2p + 3q = 18$ and $4p^2 + 4pq - 3q^2 - 36 = 0$, then what is $(2p + q)$ equal to?
 (a) 6 (b) 7
 (c) 10 (d) 20
[NDA (I) 2015]
53. In solving a problem that reduces to a quadratic equation, one student makes a mistake in the constant term and obtain 8 and 2 for roots. Another student makes a mistake only in the coefficient of first degree term and finds -9 and -1 for roots. The correct equation is:
 (a) $x^2 - 10x + 9 = 0$ (b) $x^2 + 10x + 9 = 0$
 (c) $x^2 - 10x + 6 = 0$ (d) $x^2 - 8x - 9 = 0$
[NDA (I) 2015]
54. If m and n are roots of the equation $(x + p)(x + q) - k = 0$, then roots of the equation: $(x - m)(x - n) + k = 0$ are
 (a) p and q (b) $1/p$ and $1/q$
 (c) $-p$ and $-q$ (d) $p + q$ and $p - q$
[NDA (I) 2015]
55. Consider the following statements in respect of the given equation.
 $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$
 I. All the roots of the equation are complex.
 II. The sum of all the roots of the equation is 6.
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (I) 2015]
56. If the sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to the sum of their squares, then:
 (a) $a^2 + b^2 = c^2$ (b) $a^2 + b^2 = a + b$
 (c) $ab + b^2 = 2ac$ (d) $ab - b^2 = 2ac$
[NDA (II) 2015]
57. If the roots of the equation $x^2 - nx + m = 0$ differ by 1, then:
 (a) $n^2 - 4m - 1 = 0$ (b) $n^2 + 4m - 1 = 0$
 (c) $m^2 + 4n + 1 = 0$ (d) $m^2 - 4n - 1 = 0$
[NDA (II) 2015]
58. The number of real roots of the equation $x^2 - 3|x| + 2 = 0$ is:
 (a) 4 (b) 3
 (c) 2 (d) 1
[NDA (II) 2015]
- Direction (for next two):**
 Let α and β ($\alpha < \beta$) be the roots of the equation $x^2 + bx + c = 0$, where $b > 0$ and $c < 0$.
59. Consider the following:
 I. $\beta < -\alpha$ II. $\beta < |\alpha|$
 Which of the above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (I) 2016]
60. Consider the following:
 I. $\alpha + \beta + \alpha\beta > 0$ II. $\alpha^2\beta + \beta^2\alpha > 0$
 Which of the above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (I) 2016]
61. If one root of the equation: $(l - m)x^2 + lx + 1 = 0$ is double the other and l is real, then what is the greatest value of m ?
 (a) $-9/8$ (b) $9/8$
 (c) $-8/9$ (d) $8/9$
[NDA (I) 2016]
- Direction (for next two):**
 Given that $\tan\alpha$ and $\tan\beta$ are the roots of the equation $x^2 + bx + c = 0$ with $b \neq 0$
62. What is $\tan(\alpha + \beta)$ equal to?
 (a) $b(c - 1)$ (b) $c(b - 1)$
 (c) $c(b - 1)^{-1}$ (d) $b(c - 1)^{-1}$
[NDA (I) 2016]
63. What is $\sin(\alpha + \beta) \sec\alpha \sec\beta$ equal to?
 (a) b (b) $-b$
 (c) c (d) $-c$
[NDA (I) 2016]
64. If $x^2 - px + 4 > 0$ for all real values of x , then which one of the following is correct?
 (a) $|p| < 4$ (b) $|p| \leq 4$
 (c) $|p| > 4$ (d) $|p| \geq 4$
[NDA (I) 2016]
65. If both the roots of the equation $x^2 - 2kx + k^2 - 4 = 0$ lie between -3 and 5 , then which one of the following is correct?
 (a) $-2 < k < 2$ (b) $-5 < k < 3$
 (c) $-3 < k < 5$ (d) $-1 < k < 3$
[NDA (II) 2016]
- Direction (for next two):** Consider the following the next two items that follow: Let α and β be the roots of the equation. $x^2 - (1 - 2a^2)x + (1 - 2a^2) = 0$
66. Under what condition does the above equation have real roots?
 (a) $a^2 < 1/2$ (b) $a^2 > 1/2$

- (c) $a^2 \leq 1/2$ (d) $a^2 \geq 1/2$ [NDA (II) 2016]
67. Under what condition is $\frac{1}{\alpha^2} + \frac{1}{\beta^2} < 1$?
- (a) $a^2 < 1/2$ (b) $a^2 > 1/2$
 (c) $a^2 > 1$ (d) $a^2 \in \left(\frac{1}{3}, \frac{1}{2}\right)$ [NDA (II) 2016]
- Direction (for next two):** Consider the following for the next two items that follow:
 $2x^2 + 3x - \alpha = 0$ has roots -2 and β while the equation $x^2 - 3mx + 2m^2 = 0$ has both roots positive, where $\alpha > 0$ and $\beta > 0$.
68. What is the value of α ?
- (a) $\frac{1}{2}$ (b) 1
 (c) 2 (d) 4 [NDA (II) 2016]
69. If $\beta, 2, 2m$ are in GP, then what is the value of $\beta \sqrt{m}$?
- (a) 1 (b) 2
 (c) 4 (d) 6 [NDA (II) 2016]
70. If $c > 0$ and $4a + c < 2b$, then $ax^2 - bx + c = 0$ has a root in which one of the following intervals?
- (a) (0,2) (b) (2,3)
 (c) (3,4) (d) (-2,0) [NDA (II) 2016]
71. If the difference between the roots of the equation $x^2 + kx + 1 = 0$ is strictly less than $\sqrt{5}$, where $|k| \geq 2$, then k can be any element of the interval:
- (a) $(-3, -2) \cup (2, 3)$ (b) $(-3, 3)$
 (c) $(-3, -2] \cup [2, 3)$ (d) None of these [NDA (I) 2017]
72. If the roots of the equation $x^2 + px + q = 0$ are in the same ratio as those of the equation $x^2 + lx + m = 0$, then which one of the following is correct?
- (a) $p^2m = l^2q$ (b) $m^2p = l^2q$
 (c) $m^2p = q^2l$ (d) $m^2p^2 = l^2q$ [NDA (I) 2017]
73. If $\cot \alpha$ and $\cot \beta$ are the roots of the equation $x^2 + bx + c = 0$ with $b \neq 0$, then the value of $\cot(\alpha + \beta)$ is:
- (a) $\frac{c-1}{b}$ (b) $\frac{1-c}{b}$
 (c) $\frac{c}{c-1}$ (d) $\frac{b}{1-c}$ [NDA (I) 2017]
74. If the graph a quadratic polynomial lies entirely above x-axis, which one of the following is correct?
- (a) Both the roots are real
 (b) One root is real and the other is complex
 (c) Both the roots are complex
 (d) Cannot say [NDA (I) 2017]
75. The roots of the equation $(q-r)x^2 + (r-p)x + (p-q) = 0$ are:
- (a) $\frac{(r-p)}{(q-r)}, \frac{1}{2}$ (b) $\frac{(p-q)}{(q-r)}, 1$
 (c) $\frac{(q-r)}{(p-q)}, 1$ (d) $\frac{(r-p)}{(p-q)}, \frac{1}{2}$ [NDA (II) 2017]

76. The sum of all real roots of the equation $|x-3|^2 + |x-3| - 2 = 0$ is:
- (a) 2 (b) 3
 (c) 4 (d) 6 [NDA (II) 2017]
77. If α and β are the roots of the equation $3x^2 + 2x + 1 = 0$, then the equation whose roots are $\alpha + \beta^{-1}$ and $\beta + \alpha^{-1}$ is:
- (a) $3x^2 + 8x + 16 = 0$ (b) $3x^2 - 8x - 16 = 0$
 (c) $3x^2 + 8x - 16 = 0$ (d) $x^2 + 8x + 16 = 0$ [NDA (II) 2017]
78. In ΔPQR , $\angle R = \frac{\pi}{2}$. If $\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of the equation $ax^2 + bx + c = 0$, then which one of the following is correct?
- (a) $a = b + c$ (b) $b = c + a$
 (c) $c = a + b$ (d) $b = c$ [NDA (II) 2017]
79. It is given that the roots of the equation $x^2 - 4x - \log_3 P = 0$ are real. For this, the minimum value of P is:
- (a) $1/27$ (b) $1/64$
 (c) $1/81$ (d) 1 [NDA-2017(2)]
80. The equation $|1-x| + x^2 = 5$ has:
- (a) A rational root and an irrational root
 (b) Two rational roots
 (c) Two irrational roots
 (d) No real roots [NDA (I) 2018]
81. Suppose $f(x)$ is such a quadratic expression that it is positive for all real x .
 If $g(x) = f(x) + f'(x) + f''(x)$, then for any real x .
- (a) $g(x) < 0$ (b) $g(x) > 0$
 (c) $g(x) = 0$ (d) $g(x) \geq 0$ [NDA-2018(2)]
82. The ratio of roots of the equations $ax^2 + bx + c = 0$ and $px^2 + qx + r = 0$ are equal. If D_1 and D_2 are respective discriminates, then what is $\frac{D_1}{D_2}$ equal to?
- (a) $\frac{a^2}{p^2}$ (b) $\frac{b^2}{q^2}$
 (c) $\frac{c^2}{r^2}$ (d) None of these [NDA (II) 2018]
83. If $\cos \alpha$ and $\cos \beta$ ($0 < \alpha < \beta < \pi$) are the roots of the quadratic equation $4x^2 - 3 = 0$, then what is the value of $\sec \alpha \times \sec \beta$?
- (a) $-\frac{4}{3}$ (b) $\frac{4}{3}$
 (c) $\frac{3}{4}$ (d) $\left(-\frac{3}{4}\right)$ [NDA (II) 2018]
84. If α and β ($\neq 0$) are the roots of the quadratic equation $x^2 + \alpha x - \beta = 0$, then the quadratic expression $-x^2 + \alpha x + \beta$ where $x \in \mathbb{R}$ has:
- (a) Least value $-\frac{1}{4}$ (b) Least value $\frac{9}{4}$
 (c) Greatest value $\frac{1}{4}$ (d) Greatest value $\frac{9}{4}$ [NDA (II) 2018]
85. Consider the following expression:

1. $x + x^2 - \frac{1}{x}$

2. $\sqrt{ax^2 + bx + x - c + \frac{d}{x} - \frac{e}{x^2}}$

3. $3x^2 - 5x + ab$

4. $\frac{2}{x^2 - ax + b^3}$

5. $\frac{1}{x} - \frac{2}{x+5}$

Which of the above are rational expressions.

- (a) 1, 4 and 5 only (b) 1, 3, 4 and 5 only
(c) 2, 4 and 5 only (d) 1 and 2 only

[NDA (II) 2018]

86. What are the roots of the equation $|x^2 - x - 6| = x + 2$?

- (a) -2, 1, 4 (b) 0, 2, 4
(c) 0, 1, 4 (d) -2, 2, 4

[NDA (I) 2019]

87. If a and b are the different real roots of the equation $px^2 + qx + r = 0$ (where p, q, r are positive), then which of the following is correct?

- (a) $a > 0, b > 0$ (b) $a < 0, b < 0$
(c) $a > 0, b < 0$ (d) $a < 0, b > 0$

[NDA (I) 2019]

88. If roots of the equation $x^2 + px + q = 0$ are $\tan 19^\circ$ and $\tan 26^\circ$, then which of the following is correct?

- (a) $q - p = 1$ (b) $p - q = 1$
(c) $p + q = 2$ (d) $p + q = 3$

[NDA (I) - 2019]

89. Number of real roots of the equation $x^2 + 9|x| + 20 = 0$ is/are

- (a) zero (b) one
(c) two (d) three

[NDA (I) 2019]

90. What is the solution of $x \leq 4, y \geq 0$ and $x \leq -4, y \leq 0$?

- (a) $x \geq -4, y \leq 0$ (b) $x \leq 4, y \geq 0$
(c) $x \leq -4, y = 0$ (d) $x \geq -4, y = 0$

[NDA (II) 2019]

91. How many real roots of the equation $x^2 + 3|x| + 2 = 0$ have

- (a) zero (b) one
(c) two (d) four

[NDA (II) 2019]

92. Consider the following statements in respect of the quadratic equation $4(x - p)(x - q) - r^2 = 0$ where p, q, r are real numbers

1. The roots are real
2. The roots are equal if $p = q$ and $r = 0$

Which of the above statements is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) 2019]

93. If both p and q belongs to set $\{1, 2, 3, 4\}$, then how many equations of the form $px^2 + qx + 1 = 0$ will have real roots?

- (a) 12 (b) 10
(c) 7 (d) 6

[NDA (II) 2019]

94. If the roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ are equal, then which one of the following is correct?

- (a) a, b, c are in AP
(b) a, b, c are in GP
(c) a, b, c are in HP

(d) a, b, c do not follow any regular pattern

[NDA (II) 2019]

95. If $|x^2 - 3x + 2| > x^2 - 3x + 2$, then which one of the following is correct?

- (a) $x \leq 1$ or $x \geq 2$
(b) $1 \leq x \leq 2$
(c) $1 < x < 2$
(d) x is any real value except 3 and 4

[NDA (II) 2019]

96. Under which one of the following conditions will the quadratic equation $x^2 + mx + 2 = 0$ always have real roots?

- (a) $2\sqrt{3} \leq m^2 < 8$ (b) $\sqrt{3} \leq m^2 < 4$
(c) $m^2 \geq 8$ (d) $m^2 \leq \sqrt{3}$

[NDA (II) 2019]

97. What is the value of

$$2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots}}}$$

- (a) $\sqrt{2} - 1$ (b) $\sqrt{2} + 1$
(c) 3 (d) 4

[NDA (II) 2019]

98. Let $A \cup B = \{x | (x - a)(x - b) > 0, \text{ where } a < b\}$, what are A and B equal to?

- (a) $A = \{x | x > a\}$ and $B = \{x | x > b\}$
(b) $A = \{x | x < a\}$ and $B = \{x | x > b\}$
(c) $A = \{x | x < a\}$ and $B = \{x | x < b\}$
(d) $A = \{x | x > a\}$ and $B = \{x | x < b\}$

[NDA (II) 2019]

99. If p and q are the roots of the equation $x^2 - 30x + 221 = 0$, what is the value of $p^3 + q^3$?

- (a) 7010 (b) 7110
(c) 7210 (d) 7240

[NDA (II) 2019]

100. The roots α and β of a quadratic equation satisfy the relations $\alpha + \beta = \alpha^2 + \beta^2$ and $\alpha\beta = \alpha^2\beta^2$. What is number of such quadratic equations?

- (a) 0 (b) 2
(c) 3 (d) 4

[NDA 2020]

101. If $1.5 \leq x \leq 4.5$, then which one of the following is correct?

- (a) $(2x - 3)(2x - 9) > 0$ (b) $(2x - 3)(2x - 9) < 0$
(c) $(2x - 3)(2x - 9) \geq 0$ (d) $(2x - 3)(2x - 9) \leq 0$

[NDA 2020]

102. If α and β are the roots of the equation $4x^2 + 2x - 1 = 0$, then which one of the following is correct?

- (a) $\beta = -2\alpha^2 - 2\alpha$ (b) $\beta = 4\alpha^2 - 3\alpha$
(c) $\beta = \alpha^2 - 3\alpha$ (d) $\beta = -2\alpha^2 + 2\alpha$

[NDA (I) 2021]

103. If the roots of the quadratic equation $x^2 + 2x + k = 0$ are real, then

- (a) $k < 0$ (b) $k \leq 0$
(c) $k < 1$ (d) $k \leq 1$

[NDA (I) 2021]

104. If one root of $5x^2 + 26x + k = 0$ is reciprocal of other then what is the value of k?

- (a) 2 (b) 3
(c) 5 (d) 8

[NDA (I) 2021]

105. If the roots of the equation $4x^2 - (5k + 1)x + 5k = 0$ differ by unity, then which one of the following is a possible value of k?

- (a) -3 (b) -1

$$(c) -\frac{1}{5}$$

$$(d) -\frac{3}{5}$$

[NDA (II) 2021]

106. The quadratic equation $3x^2 - (k^2 + 5k)x + 3k^2 - 5k = 0$ has real roots of equal magnitude and opposite sign. Which one of the following is correct?

$$(a) 0 < k < \frac{5}{3}$$

$$(b) 0 < k < \frac{3}{5}$$

$$(c) \frac{3}{5} < k < \frac{5}{3}$$

$$(d) \text{ No such value of } k \text{ exists}$$

[NDA (II) 2021]

107. If p and q are the non-zero roots of the equation $x^2 + px + q = 0$, then how many possible values can q have?

(a) Nil

(b) One

(c) Two

(d) Three

[NDA (II) 2021]

108. Consider all the real roots of the equation $x^4 - 10x^2 + 9 = 0$. What is the sum of the absolute values of the roots?

(a) 4

(b) 6

(c) 8

(d) 10

[NDA (II) 2021]

109. Consider the in equations $5x - 4y + 12 < 0$, $x + y < 2$, $x < 0$ and $y > 0$. Which one of the following points lies in the common region?

(a) (0, 0)

(b) (-2, 4)

(c) (-1, 4)

(d) (-1, 2)

[NDA (I) 2022]

110. Let α and β be the roots of the equation $x^2 + px + q = 0$. If α^3 and β^3 are the roots of the equation $x^2 + mx + n = 0$, then what is the value of $m + n$?

$$(a) p^3 + q^3 + pq$$

$$(b) p^3 + q^3 - pq$$

$$(c) p^3 + q^3 + 3pq$$

$$(d) p^3 + q^3 - 3pq$$

[NDA (I) 2022]

111. Let α and β be the roots of the equation $x^2 - ax - bx + ab - c = 0$. What is the quadratic equation whose roots are a and b ?

$$(a) x^2 - \alpha x - \beta x + \alpha\beta + c = 0$$

$$(b) x^2 - \alpha x - \beta x + \alpha\beta - c = 0$$

$$(c) x^2 + \alpha x + \beta x + \alpha\beta + c = 0$$

$$(d) x^2 + \alpha x + \beta x + \alpha\beta - c = 0$$

[NDA (I) 2022]

112. If the roots of the equation $x^2 - ax - bx - cx + bc + ca = 0$ are equal, then which one of the following is correct?

$$(a) a + b + c = 0$$

$$(b) a - b + c = 0$$

$$(c) a + b - c = 0$$

$$(d) -a + b + c = 0$$

[NDA (I) 2022]

113. Let α and β ($\alpha > \beta$) be the roots of the equation $x^2 - 8x + q = 0$. If $\alpha^2 - \beta^2 = 16$, then what is the value of q ?

(a) -15

(b) -10

(c) 10

(d) 15

[NDA (I) 2022]

114. For how many quadratic equations, the sum of roots is equal to the product of roots?

(a) 0

(b) 1

(c) 3

(d) Infinitely many

[NDA 2022 (II)]

115. Let p, q ($p > q$) be the roots of the quadratic equation $x^2 + bx + c = 0$ where $c > 0$. If $p^2 + q^2 - 11pq = 0$, then what is $p - q$ equal to?

$$(a) 3\sqrt{c}$$

$$(b) 3c$$

$$(c) 9\sqrt{c}$$

$$(d) 9c$$

[NDA 2022 (II)]

116. How many real numbers satisfy the equation $|x - 4| + |x - 7| = 15$?

(a) only one

(b) only two

(c) only three

(d) infinitely many

[NDA - 2023 (1)]

117. α and β are distinct real roots of the quadratic equation $x^2 + ax + b = 0$. Which of the following statements is/are sufficient to find α ?

$$1. \alpha + \beta = 0, \alpha^2 + \beta^2 = 2$$

$$2. \alpha\beta^2 = -1, a = 0$$

Select the correct answer using the code given below:

(a) 1 only

(b) 2 only

(c) both 1 and 2

(d) Neither 1 nor 2

[NDA - 2023 (1)]

118. If $2 - i\sqrt{3}$ where $i = \sqrt{-1}$ is a root of the equation $x^2 + ax + b = 0$, then what is the value of $(a + b)$?

(a) -11

(b) -3

(c) 0

(d) 3

[NDA - 2023 (1)]

119. For how many integral values of k , the equation $x^2 - 4x + k = 0$, where k is an integer has real roots and both of them lie in the interval $(0, 5)$?

(a) 3

(b) 4

(c) 5

(d) 6

[NDA - 2023 (1)]

Consider the following for the next (02) items that follow:

Consider the equation $(1-x)^4 + (5-x)^4 = 82$

120. What is the number of real roots of the equation?

(a) 0

(b) 2

(c) 4

(d) 8

[NDA-2023 (2)]

121. What is the sum of all the roots of the equation?

(a) 24

(b) 12

(c) 10

(d) 6

[NDA-2023 (2)]

Consider the following for the next (02) items that follow:

A quadratic equation is given by $(a + b)x^2 - (a + b + c)x + k = 0$, where a, b, c are real.

122. If $k = \frac{c}{2}$, ($c \neq 0$), then the roots of the equation are:

(a) Real and equal

(b) Real and unequal

(c) Real iff $a > c$

(d) Complex but not real

[NDA-2023 (2)]

123. If $k = c$, then the roots of the equation are:

$$(a) \frac{a+c}{a+b} \text{ and } \frac{b}{a+b}$$

$$(b) \frac{a+c}{a+b} \text{ and } -\frac{b}{a+b}$$

$$(c) 1 \text{ and } \frac{c}{a+b}$$

$$(d) -1 \text{ and } -\frac{c}{a+b}$$

[NDA-2023 (2)]

124. If $-\sqrt{2}$ and $\sqrt{3}$ are roots of the equation $a_0 + a_1x + a_2x^2 + a_3x^3 + x^4 = 0$ where a_0, a_1, a_2, a_3 are integers, then which one of the following is correct?

$$(a) a_2 = a_3 = 0$$

$$(b) a_2 = 0 \text{ and } a_3 = -5$$

$$(c) a_0 = 6, a_3 = 0$$

$$(d) a_1 = 0 \text{ and } a_2 = 5$$

[NDA-2024 (1)]

125. Under which one of the following conditions does the equation $(\cos \beta - 1)x^2 + (\cos \beta)x + \sin \beta = 0$ in x have a real root for $\beta \in [0, \pi]$?

$$(a) 1 - \cos \beta < 0$$

$$(b) 1 - \cos \beta \leq 0$$

$$(c) 1 - \cos \beta > 0$$

$$(d) 1 - \cos \beta \geq 0$$

[NDA-2024 (1)]

126. If a , b and c ($a > 0$, $c > 0$) are in GP, then consider the following in respect of the equation $ax^2 + bx + c = 0$:
1. The equation has imaginary roots
 2. The ratio of the roots of the equation is $1 : \omega$ where ω is a cube root of unity
 3. The product of roots of the equation is $\left(\frac{b^2}{a^2}\right)$.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2024 (1)]

127. If $x^2 + mx + n$ is an integer for all integral values of x , then which of the following is/are correct?

1. m must be an integer

2. n must be an integer

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below

The roots of the quadratic equation

$a^2(b^2 - c^2)x^2 + b^2(c^2 - a^2)x + c^2(a^2 - b^2) = 0$ are equal. Given that $(a^2 \neq b^2 \neq c^2)$

128. Which one of the following statement is correct?

- (a) a^2, b^2, c^2 are in AP
(b) a^2, b^2, c^2 are in GP
(c) a^2, b^2, c^2 are in HP
(d) a^2, b^2, c^2 are neither in AP nor in GP nor in HP

[NDA-2024 (2)]

129. Which one of the following is a root of the equation?

- (a) $\frac{b^2(c^2 - a^2)}{a^2(c^2 - b^2)}$ (b) $\frac{b^2(c^2 - a^2)}{a^2(b^2 - c^2)}$
(c) $\frac{b^2(c^2 - a^2)}{2a^2(c^2 - b^2)}$ (d) $\frac{b^2(c^2 - a^2)}{2a^2(b^2 - c^2)}$

[NDA-2024 (2)]

130. What is the number of real roots of the equation

$$(x-1)^2 + (x-3)^2 + (x-5)^2 = 0$$

- (a) none (b) only one
(c) only two (d) Three

[NDA-2024 (2)]

131. If n is a root of the equation $x^2 + px + m = 0$ and m is a root of the equation $x^2 + px + n = 0$, where $m \neq n$, then what is the value of $p + m + n$?

- (a) -1 (b) 0
(c) 1 (d) 2

[NDA-2024 (2)]

132. If $f(x) = 9x - 8\sqrt{x}$ such that $g(x) = f(x) - 1$, then which one of the following is correct?

- (a) $g(x)$ has no real roots
(b) $g(x) = 0$ has only one real root which is an integer
(c) $g(x) = 0$ has two real roots which are integers.
(d) $g(x) = 0$ has only one real root which is not an integer.

[NDA-2024 (2)]

133. If one root of the equation $x^2 - kx + k = 0$ exceeds the other by $2\sqrt{3}$, then which one of the following is a value of k ?

- (a) 3 (b) 6
(c) 9 (d) 12

[NDA-2025 (1)]

134. If $x + \frac{5}{y} = 4$ and $y + \frac{5}{x} = -4$, then what is $(x + y)$ equal to?

- (a) 0 (b) 1
(c) 2 (d) -5

[NDA-2025 (1)]

Direction for next two:

Let α and β be the roots of the quadratic equation

$$x^2 + (\log_{0.5}(a^2))x + (\log_{0.5}(a^2))^4 = 0$$

Where $a^2 \neq 1$ and $\log_{0.5}(a^2) > 0$.

Further, $\beta^2 = \alpha(\log_{a^2}(0.5))$

[NDA-2025 (2)]

135. What is β equal to?

- (a) $\log_{a^2}(0.5)$ (b) $\log_{0.5}(a^2)$
(c) $2\log_{a^2}(0.5)$ (d) $2\log_{0.5}(a^2)$

136. What is the relation between α and β ?

- (a) $\alpha = 2\beta$ (b) $2\alpha = \beta$
(c) $\alpha = -2\beta$ (d) $2\alpha = -\beta$

Direction for next two:

Consider the equation

$$abx^2 + bcx + ca = cax^2 + abx + bc$$

[NDA-2025 (2)]

137. If the roots of the equation are equal, then which one of the following is correct?

- (a) $ac = b^2$ (b) $a + c = 2b$
(c) $\frac{1}{a} + \frac{1}{c} = \frac{1}{2b}$ (d) $\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$

138. If the roots of the equation are equal, then a , b , c are in

- (a) AP (b) GP
(c) HP (d) None of these

ANSWER KEY

1.	d	2.	a	3.	a	4.	d	5.	c	6.	b	7.	a	8.	d	9.	c	10.	c
11.	d	12.	c	13.	d	14.	c	15.	a	16.	d	17.	c	18.	d	19.	c	20.	c
21.	d	22.	a	23.	a	24.	a	25.	a	26.	a	27.	d	28.	b	29.	c	30.	c
31.	a	32.	b	33.	a	34.	d	35.	b	36.	c	37.	d	38.	c	39.	d	40.	c
41.	d	42.	b	43.	c	44.	a	45.	d	46.	a	47.	d	48.	c	49.	c	50.	d
51.	a	52.	c	53.	a	54.	c	55.	b	56.	c	57.	a	58.	a	59.	c	60.	b
61.	b	62.	d	63.	b	64.	a	65.	d	66.	d	67.	a	68.	c	69.	a	70.	a
71.	c	72.	a	73.	b	74.	c	75.	b	76.	d	77.	a	78.	c	79.	c	80.	a
81.	b	82.	b	83.	a	84.	d	85.	b	86.	d	87.	b	88.	a	89.	a	90.	c
91.	a	92.	c	93.	c	94.	c	95.	c	96.	c	97.	b	98.	b	99.	b	100.	d
101.	d	102.	a	103.	d	104.	c	105.	c	106.	d	107.	b	108.	c	109.	d	110.	d
111.	a	112.	c	113.	d	114.	d	115.	a	116.	b	117.	c	118.	d	119.	b	120.	b
121.	b	122.	b	123.	c	124.	c	125.	d	126.	d	127.	c	128.	c	129.	c	130.	a
131.	c	132.	b	133.	b	134.	a	135.	b	136.	c	137.	d	138.	c				

Solutions

Sol. 1. (d)

$$\alpha\beta < 0$$

$$\frac{c}{a} < 0$$

The condition that both roots are of opposite sign is, $a < 0, c > 0$.

Sol. 2. (a)

$$(2-\sqrt{3})x^2 - (7-4\sqrt{3})x + (2+\sqrt{3}) = 0$$

$$\alpha + \beta = \frac{7-4\sqrt{3}}{2-\sqrt{3}} = (7-4\sqrt{3})(2+\sqrt{3})$$

$$= 14 - 12 - 8\sqrt{3} + 7\sqrt{3} = 2 - \sqrt{3}$$

Sol. 3. (a)

If one root of the equation $ax^2 + bx + c = 0$, $a \neq 0$ is reciprocal of the other root

$$\text{Then } \alpha = \frac{1}{\beta}$$

$$\text{We know that } \alpha\beta = \frac{c}{a}$$

$$\Rightarrow \frac{1}{\beta} \cdot \beta = \frac{c}{a} \Rightarrow c = a$$

Sol. 4. (d)

Let the root of the quadratic equation is α, β

$$\therefore \alpha + \beta = -k, \alpha\beta = -b$$

$$\text{Now, } \alpha^2 + \beta^2 = 2b \quad (\text{given})$$

$$(\alpha + \beta)^2 - 2\alpha\beta = 2b \Rightarrow k^2 + 2b = 2b$$

$$\therefore k = 0$$

Sol. 5. (c)

$$9 - 24 + k = 0$$

$$k = 15$$

Sol. 6. (b)

Given quadratic equation

$$ax^2 + bx + c = 0, a \neq 0$$

Let the roots of equation be (α, β) ,

$$\text{Where } \beta = 2\alpha$$

$$\text{Then, sum of roots} = -b/a$$

$$\Rightarrow \alpha + 2\alpha = -b/a$$

$$\Rightarrow \alpha = -b/3a \quad \dots(i)$$

Product of roots $= c/a$

$$\Rightarrow 2\alpha^2 = c/a$$

$$\Rightarrow 2\left(-\frac{b}{3a}\right)^2 = \frac{c}{a} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 2b^2 = 9a^2 \cdot c/a$$

$$\therefore 2b^2 = 9ac$$

Sol. 7. (a)

The given equation is, $x^4 - 26x^2 + 25 = 0$

$$\Rightarrow x^2 - 25x^2 - x^2 + 25 = 0$$

$$\Rightarrow x^2(x^2 - 25) - 1(x^2 - 25) = 0$$

$$\Rightarrow (x^2 - 25)(x^2 - 1) = 0$$

$$\Rightarrow (x - 5)(x + 5)(x - 1)(x + 1) = 0$$

$$\therefore x = -5, -1, 1, 5$$

So, the solution set is $\{-5, -1, 1, 5\}$

Sol. 8. (d)

The given equation $x^2 - 2px + p^2 - q^2 + 2qr - r^2 = 0$ and p, q, r are rational numbers.

$$\text{Now, } D = B^2 - 4AC$$

$$D = 4p^2 - 4\{p^2 - (q-r)^2\}$$

$$D = 4p^2 - 4p^2 + 4(q-r)^2$$

$$D = 4(q-r)^2 = \text{rational and positive}$$

So, the roots of the equation will always be rational.

Sol. 9. (c)

$$x + y \leq 4$$

(x, y) can be

$$\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)\}$$

required number of ordered pair = 6

Sol. 10. (c)

given α and β are the roots of equation $4x^2 + 3x + 7 = 0$, so $\alpha + \beta = -3/4$ and $\alpha\beta = 7/4$.

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$\frac{9}{16} - \frac{7}{2} = -\frac{47}{49}$$

Sol. 11 (d) .

$$y = \sqrt{8 + 2\sqrt{8 + 2\sqrt{8 + 2\sqrt{8 + \dots}}}} \quad \dots(i)$$

Squaring both the sides, we get

$$y^2 = 8 + 2\sqrt{8 + 2\sqrt{8 + 2\sqrt{8 + \dots}}}$$

$$\text{or } y^2 = 8 + 2y \quad [\text{from (i)}]$$

$$\text{or } y^2 - 2y - 8 = 0$$

$$\Rightarrow y = 4 \text{ or } -2$$

Since the value of y cannot be $-ve$, therefore $y = 4$

Sol. 12.(c)

Given, quadratic equation is;

$$(x-a)(x-b) = c, c \neq 0$$

$$\Rightarrow x^2 - (a+b)x + (ab-c) = 0$$

The roots of this equation are (α, β)

$$\text{Then, } \alpha + \beta = -(a+b) = a+b \quad \dots(i)$$

$$\text{and } \alpha\beta = ab - c \quad \dots(ii)$$

Sol. 13 (d)

We know, that, if one root of the quadratic equation is complex, then its other root is its conjugate.

Now, $x^2 - 4x + 29 = 0$ has one root $\alpha = 2+5i$ so, $2-5i$ is the second root of this equation.

Sol. 14 (c)

Given quadratic equation is:

$$a(b-c)x^2 + b(c-a)x + c(a-b) = 0$$

one root of above equation is 1

$$\therefore \text{Product of the roots} = \frac{c(a-b)}{a(b-c)}$$

$$\text{so other root will be } \Rightarrow \alpha = \frac{c(a-b)}{a(b-c)}$$

Sol. 15(a)

The given quadratic equation is

$$2(y+2)^2 - 5(y+2) = 12$$

$$\Rightarrow 2(y+2)^2 - 5(y+2) - 12 = 0$$

$$\text{Let } z = y + 2 \quad \dots(i)$$

$$\Rightarrow 2z^2 - 5z - 12 = 0$$

$$\Rightarrow 2z^2 - 8z + 3z - 12 = 0$$

$$\Rightarrow (2z + 3)(z - 4) = 0$$

$$z = -\frac{3}{2}, 4$$

$$\Rightarrow y + 2 = -\frac{3}{2}, 4 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow y = -2 - \frac{3}{2}, 2 \Rightarrow y = -\frac{7}{2}, 2$$

Sol. 16(d) Let the roots of the equation, $3x^2 + 5x + q = 0$ is (α, α)

$$\text{Then, sum of the roots} = -\left(\frac{-5}{3}\right)$$

$$\Rightarrow \alpha + \alpha = -\frac{5}{3} \Rightarrow \alpha = -\frac{5}{6}$$

The root satisfies the given equation

$$\Rightarrow 3\alpha^2 = 75 \Rightarrow \alpha = 25/12$$

Sol. 17(c)

Let the roots of the equation $x^2 - px + q = 0$ is (α, β) and the roots of the equation $x^2 - ax + b = 0$ is (α, α)

$$\text{Then, } \alpha + \beta = p \quad \text{and } \alpha\beta = q \dots (i)$$

$$\alpha + \alpha = a \quad \text{and } \alpha \cdot \alpha = b$$

$$\Rightarrow \alpha = \frac{a}{2} \quad \text{and } \alpha^2 = b$$

$$\Rightarrow \left(\frac{a}{2}\right)^2 = b \Rightarrow a = a^2 = 4b$$

From Eq. (i), $\alpha\beta = q$

$$\Rightarrow \frac{a}{2} \cdot \beta = q \Rightarrow \beta = \frac{2q}{a}$$

Now, putting the value of α and β in the following equation

$$\alpha + \beta = p \Rightarrow \frac{a}{2} + \frac{2q}{a} = p \Rightarrow a^2 + 4q = 2ap$$

$$\Rightarrow 4b + 4q = 2ap \quad [\text{from Eq. (ii)}]$$

$$\therefore 2(b+q) = ap$$

Sol. 18(d)

for real root $b^2 - 4ac \geq 0$

$$\Rightarrow 16 + 4 \log_3 P \geq 0$$

$$\Rightarrow 4 \log_3 P \geq -16$$

$$\Rightarrow \log_3 P \geq -4$$

$$\Rightarrow P \geq 1/81$$

Sol. 19(c)

Given equation is

$$\tan^4 x - 2\sec^2 x + a^2 = 0$$

$$(\tan^2 x)^2 - 2(1 + \tan^2 x) + a^2 = 0$$

$$(\tan^2 x)^2 - 2\tan^2 x + a^2 - 2 = 0$$

This is the quadratic equation in $\tan x$

so roots of this will be real

$$\therefore b^2 - 4ac \geq 0$$

$$(-2)^2 - 4(a^2 - 2) \geq 0$$

$$a^2 \leq 3 \Rightarrow |a| \leq \sqrt{3}$$

Sol. 20(c)

$$x^2 - x \sin \theta + a = 0$$

$$\text{discriminant} \geq 0$$

$$\sin^2 \theta - 4a \geq 0$$

$$a \leq \frac{\sin^2 \theta}{4} \Rightarrow a \leq 1/4$$

Sol. 21(d)

Given, quadratic equation $x^2 + 2x - 143 = 0$

Let (α, β) be the roots of this equation

$$\text{Then, } \alpha + \beta = -2 \text{ and } \alpha\beta = -143$$

We have,

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-2)^2 - 2(-143)$$

$$= 4 + 286 = 290$$

Sol. 22 (b)

Since, one root of $x^2 + ax - b = 0$ is 1

$$\therefore 1^2 + 1 \cdot a - b = 0 \Rightarrow 1 + a - b = 0$$

$$\Rightarrow a - b = -1$$

Sol. 23 (a)

If α and β are the roots of the equation

$$x^2 - q(1+x) - r = 0$$

then

$$\alpha + \beta = q \text{ and } \alpha\beta = -q - r$$

$$(1+\alpha)(1+\beta) = (1+\alpha+\beta+\alpha\beta)$$

$$= 1 + q - q - r = 1 - r$$

Sol. 24(a)

The solution of the simultaneous linear equations

$2x + y = 6$ and $3y = 8 + 4x$ is $(1, 4)$ which satisfy our first option

$$x + y = 5$$

Sol. 25(a)

Let the roots of the equation $ax^2 + bx + c = 0$ are α and $(\alpha-1)$ by given condition.

Then,

$$\text{Sum of the roots} = -b/a \Rightarrow \alpha + (\alpha-1) = -b/a$$

$$\Rightarrow 2\alpha = 1 - \frac{b}{a} \Rightarrow \alpha = \frac{a-b}{2a}$$

And product of roots = c/a

$$\Rightarrow \alpha(\alpha-1) = \frac{c}{a} \Rightarrow \frac{(a-b)}{2a} \left\{ \frac{2-b}{2a} - 1 \right\} = \frac{c}{a}$$

$$\Rightarrow \frac{(a-b)}{2a} \cdot \frac{(-b-a)}{2a} = \frac{c}{a}$$

$$\Rightarrow -(a^2 - b^2) = 4ac$$

$$\Rightarrow b^2 - a^2 = 4ac \Rightarrow b^2 = a(a+4c)$$

Solution (for next two) Let α and β be the roots of the equation $ax^2 + bx + c = 0$

$$\text{Then } \alpha + \beta = -b/a$$

$$\Rightarrow 2\alpha + 2\beta = -2b/a \quad \dots (i)$$

$$\text{and } \alpha\beta = c/a$$

$$\Rightarrow (2\alpha)(2\beta) = 4c/a \quad \dots (ii)$$

Also given that, the equation $x^2 + 36x + 24 = 0$ is formed by multiplying each root of $ax^2 + bx + c = 0$ by 2

$$\therefore 2\alpha + 2\beta = -36 \quad \dots (iii)$$

$$\text{and } (2\alpha)(2\beta) = 24 \quad \dots (iv)$$

Now, from Eqs. (i) and (iv), we get

$$-36 = -2b/a \Rightarrow \dots (v)$$

From Eqs. (ii) and (iii), we get

$$24 = \frac{4c}{a} \Rightarrow \frac{c}{a} = \frac{6}{1} \quad \dots (vi)$$

Sol. 26(a)

Now, dividing Eq. (v) by Eq. (vi), we get

$$\frac{b}{c} = \frac{3}{1} \Rightarrow b : c = 3 : 1$$

Sol. 27(d)

Now, dividing Eq. (v) by Eq. (vi), we get

$$\frac{b}{a} \times \frac{3}{1} = 18 \times 6 \Rightarrow bc = 108a^2$$

Sol. 28(b)

Since, the roots of the quadratic equation $3x^2 - 5x + p = 0$ are real and unequal.

$$\therefore \text{Discriminate} > 0 \Rightarrow b^2 - 4ac > 0$$

$$\Rightarrow (-5)^2 - 4(3)(p) > 0$$

$$(\text{here, } b = -5, a = 3, c = p)$$

$$\Rightarrow 25 - 12p > 0 \Rightarrow 25 > 12p$$

$$\Rightarrow 12p < 25 \Rightarrow p < 25/12$$

Sol. 29(c)

$$\text{Sum of roots} = (m+n) + (m-n) = 2m$$

$$\text{Product of roots} = (m+n)(m-n) = m^2 - n^2$$

$\therefore$ Quadratic equation

$$x^2 - (\text{sum of roots})x + \text{Product of roots} = 0$$

$$x^2 - 2mx + (m^2 - n^2) = 0$$

Sol. 30(c)

Since, α and β are the roots of $x^2 + px - q = 0$

$$\therefore \alpha + \beta = -p, \alpha\beta = -q$$

Again, since γ, δ are the roots of

$$x^2 - px + r = 0$$

$$\therefore \gamma + \delta = p, \gamma\delta = r$$

$$(\beta + \gamma)(\beta + \delta) = \beta^2 + \beta\delta + \gamma\beta + \gamma\delta$$

$$= \beta^2 + \beta(\gamma + \delta) + \gamma\delta$$

$$= \beta^2 + \beta(p) + \gamma\delta \quad (\because \gamma + \delta = p \text{ and } \gamma\delta = r)$$

$$= \beta^2 + \beta(-\alpha - \beta) + r \quad [\because p = -(\alpha + \beta)]$$

$$= \beta^2 + (-\beta)(\alpha + \beta) + r$$

$$= \beta^2 - \alpha\beta - \beta^2 + r = -\alpha\beta + r$$

$$= -(-q) + r = q + r$$

$$\text{Hence, } (\beta + \gamma)(\beta + \delta) = q + r$$

Sol. 31(a)

Given that, α and β are the roots of the equation $x^2 + bx + c = 0$.

$$\text{Then, sum of the roots} = \alpha + \beta = \frac{-b}{1} = -b$$

$$\dots (i)$$

and product of roots

$$= \alpha \cdot \beta = \frac{c}{1} = c \quad \dots (ii)$$

$$\therefore \alpha^{-1} + \beta^{-1} = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b}{c}$$

Sol. 32(b) Given, that, $(x+1)^2 - 1 = 0$

$$\Rightarrow (x+1)^2 - (1)^2 = 0$$

$$\Rightarrow (x+1+1)(x+1-1) = 0$$

$$[\because a^2 - b^2 = (a-b)(a+b)]$$

$$\Rightarrow (x+2)(x) = 0 \therefore x = 0, -2$$

Hence, $(x+1)^2 - 1 = 0$ has two real roots.

Sol. 33(a)

$$4^x - 6 \cdot 2^x + 8 = 0$$

$$2^{2x} - 6 \cdot 2^x + 8 = 0$$

$$(2^x - 2)(2^x - 4) = 0$$

$$2^x = 2 \Rightarrow x = 1$$

$$2^x = 4 \Rightarrow x = 2$$

Sol. 34(d)

Given quadratic equation is $ax^2 + bx + c = 0$

Since, its roots are α and β

$$\therefore \text{Sum of the roots} = \alpha + \beta = -b/a$$

$$\text{and product of the roots} = \alpha\beta = c/a$$

$$\text{We have, } \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{b}{a}\right)^2 - 2 \cdot \frac{c}{a}$$

$$c/a$$

$$= \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$$

$$\text{and } \alpha^2 \beta^2 = (\alpha\beta)^2 = \left(\frac{c}{a}\right)^2 = \frac{c^2}{a^2}$$

$\therefore$ Required quadratic equation whose roots are α^2 and β^2 is

$$x^2 - (\alpha^2 + \beta^2)x + \alpha^2 \beta^2 = 0$$

$$\Rightarrow x^2 - \frac{(b^2 - 2ac)}{a^2}x + \frac{c^2}{a^2} = 0$$

$$\therefore a^2 x^2 - (b^2 - 2ac)x + c^2 = 0$$

Sol. 35(b)

Let the required quadratic equation is

$$ax^2 + bx + c = 0 \quad \dots (i)$$

$$\text{Now, sum of the roots} = \frac{-b}{a} = 3 = \frac{-(-3)}{1} \quad (\text{given})$$

$$\text{and product of the roots} = c/a = 2 = 2/1 \quad (\text{given})$$

$\therefore a = 1, b = -3$ and $c = 2$

Hence, required quadratic equation is $x^2 - 3x + 2 = 0$

Sol. 36(c)

$$\frac{1}{x-3} = \frac{1}{x+2} - \frac{1}{2}$$

By solving this we will get a quadratic equation whose degree is 2

Sol. 37(d)

Given quadratic equation $3ax^2 + 2bx + c = 0$

Let its roots be 2α and 3α .

(Since, roots are in the ratio 2:3)

$$\text{Now, sum of the roots} = 2\alpha + 3\alpha = \frac{-2b}{3a}$$

$$\Rightarrow 5\alpha = \frac{-2b}{3a} \Rightarrow \alpha = \frac{-2b}{15a} \quad \dots(i)$$

and product of the roots

$$= 2\alpha \cdot 3\alpha = \frac{c}{3a} \Rightarrow 6\alpha^2 = \frac{c}{3a} \Rightarrow \left(\frac{-2b}{15a}\right)^2 = \frac{c}{18a}$$

$$\Rightarrow \frac{4b^2}{225a^2} = \frac{c}{18a} \Rightarrow \frac{4b^2}{25a^2} = \frac{c}{2a}$$

$$\therefore 8b^2 = 25ac$$

Sol. 38(c)

The area of a rectangle whose length is five more than twice of its width is 75 square unit.

$$L \times B = 75$$

$$(2B + 5) \times B = 75$$

By solving above quadratic equation

$$B = 5 \text{ and } L = 15$$

Sol. 39(d)

Given that, the equation $x^2 + bx + 4 = 0$ have real roots, if discriminant $(D) = B^2 - 4AC \geq 0$

$$\Rightarrow b^2 - 4(1)(4) \geq 0 \Rightarrow b^2 - 16 \geq 0$$

$$\Rightarrow (b-4)(b+4) \geq 0$$

$$\therefore b \leq -4, \text{ or } b \geq 4$$

Sol. 40(c)

Given that, (α, β) are the roots of the equation $x^2 + x + 2 = 0$, then

$$\alpha + \beta = -1 \quad \dots(i)$$

$$\text{and } \alpha\beta = 2 \quad \dots(ii)$$

$$\text{Now, we have } \frac{\alpha^{10} + \beta^{10}}{\alpha^{-10} + \beta^{-10}} (\alpha\beta)^{10} = (2)^{10}$$

$$[\text{from Eq. (ii)}] \\ = 1024$$

Sol. 41(d)

Given, equation, $x^2 - 10x + 9 = 0$

Let (α, β) be the root of the given equation

$$\text{Then, } \alpha + \beta = 10 \quad \dots(i)$$

$$\text{and } \alpha\beta = 9 \quad \dots(ii)$$

Now, we use the identity

$$(\alpha, \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (10)^2 - 4(9)$$

$$= (10)^2 - 4(9)$$

$$\Rightarrow \alpha - \beta = \pm \sqrt{|\alpha - \beta|} = 8$$

Sol. 42(b)

Given quadratic equation is $ax^2 + bx + b = 0$

Let $(\alpha, \beta) = -b/a$ and $\alpha\beta = b/a$

Now, we have

$$\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} = \frac{\alpha + \beta}{\sqrt{\alpha\beta}} + \sqrt{\frac{b}{a}} = \frac{-b}{a} \times \sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}}$$

$$= -\sqrt{\frac{b}{a}} + \sqrt{\frac{b}{a}} = 0$$

Sol. 43(c)

Given equation is $x^2 - 8x + 16 = 0$

$$\Rightarrow (x - 4)^2 = 0 \Rightarrow x = 4, 4$$

$$\text{Also, discriminant} = b^2 - 4ac = 0$$

So, the roots of the equation are equal and real.

Sol. 44(a)

If one root of any quadratic equation is in the form $3a + \sqrt{b}$, then other root of this equation should

be $3a - \sqrt{b}$.

$\therefore$ Required equation is

$$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$$

$$\Rightarrow x^2 - \{(3a + \sqrt{b}) + (3a - \sqrt{b})\}x$$

$$+ \{(3a + \sqrt{b})(3a - \sqrt{b})\} = 0$$

$$\therefore x^2 - 6ax + 9a^2 - b = 0$$

Sol. 45(d)

Given quadratic equation,

$$f(x) = x^2 + 3|x| + 2 = 0$$

Case I. $f(x) = x^2 + 3x + 2 = 0$ (when, $x > 0$)

$$\Rightarrow x^2 + 2x + x + 2 = 0$$

$$\Rightarrow x(x+2) + 1(x+2) = 0$$

$$\Rightarrow (x+2)(x+1) = 0$$

$$\therefore x = -2, -1 \quad (\text{but } x > 0)$$

So, here no real roots exist.

Case II. $f(x) = x^2 - 3x + 2 = 0$ (when, $x < 0$)

$$\Rightarrow x = \frac{3 \pm \sqrt{9-8}}{2} = \frac{3+1}{2}$$

$$\Rightarrow x = 2, 1 \quad (\text{but } x < 0)$$

So, here also no real root exists

Hence, given quadratic equation has no real root.

Shortcut Method: In equation $x^2 + 3|x| + 2 = 0$, taking any value of x the LHS will always give +ve answer which can never be equal to zero. Thus, there are no real roots of the equation.

Sol. 46(a)

by solving both equations

$$x = 7 \text{ and } y = 4 \text{ then } 2x - y = 2(7) - 4 = 10.$$

Sol. 47(d)

$$\sqrt{7+4\sqrt{3}} = \sqrt{7+2\sqrt{12}} = \sqrt{(2+\sqrt{3})^2} = 2+\sqrt{3}$$

Sol. 48(c) Given that, α and β are the roots of the equation $ax^2 + bx + c = 0$, where $a \neq 0$.

Then, sum of roots $= \alpha + \beta = -b/a$

and product of roots $= \alpha\beta = c/a$

Now, we have

$$(a\alpha + b)(a\beta + b) = a^2(\alpha\beta) + ab(\alpha + \beta) + b^2$$

$$= a^2\left(\frac{c}{a}\right) + ab\left(-\frac{b}{a}\right) + b^2$$

$$= ac - b^2 + b^2 = ac.$$

Sol. 49(c)

Given equation, $2a^2x^2 - 2abx + b = 0$

When, $a < 0$ and $b > 0$

$$\therefore x = \frac{-(-2ab) \pm \sqrt{(-2ab)^2 - 4.2a^2.b}}{2.2a^2}$$

(By Quadratic formula)

$$= \frac{2ab \pm \sqrt{4a^2b^2 - 8a^2b^2}}{4ab^2}$$

$$= \frac{2ab \pm \sqrt{-4a^2b^2}}{4ab^2} = \frac{2ab \pm i2ab}{4ab^2}$$

$$= \frac{2ab(1 \pm i)}{4ab^2} = \frac{b}{2a}(1 \pm i)$$

Which shows that the roots of the given equation is always complex.

Sol. 50(d)

Every quadratic equation

$$ax^2 + bx + c = 0, \text{ where } a, b, c \in \mathbb{R}, a \neq 0$$

has at most two real roots.

Sol. 51(a)

$$\alpha + \beta = -b/a \text{ and } \alpha\beta = c/a$$

$$\text{Also, } \alpha + h + \beta + h = -q/p$$

$$\Rightarrow \alpha + \beta + 2h = -q/p$$

$$\Rightarrow 2h = -\frac{q}{p} + \frac{b}{a} \left(\therefore \frac{-b}{a} \right)$$

$$\Rightarrow h = \frac{1}{2} \left[\frac{b}{a} - \frac{q}{p} \right]$$

Sol. 52(c)

$$\text{Given, } 2p+3q = 18 \quad \dots(i)$$

$$\text{and } 4p^2 + 4pq - 3q^2 - 36 = 0$$

$$\Rightarrow (2p+3q)^2 - 8pq - 12q^2 = 36$$

$$\Rightarrow 18^2 - 4q(2p+3q) = 36 \quad [\text{from (i)}]$$

$$\Rightarrow 18^2 - 4q.18 = 36$$

$$\Rightarrow q = 4 \text{ then } p = 3$$

$$2q + p = 10.$$

Sol. 53(a)

Given, roots are 8 and 2, then the equation is

$x^2 - (\text{sum of roots})x + \text{product of roots} = 0$

$$\Rightarrow x^2 - (8+2)x + 16 = 0$$

$$\Rightarrow x^2 - 10x + 16 = \dots(i)$$

(mistake in the constant term)

For roots -9 and -1, we have

$$x^2 - (-9x - 1)x + (-9) \times (-1) = 0$$

$$\Rightarrow x^2 - 9x + 9 = 0 \quad \dots(ii)$$

(mistake in the coefficient of first degree term) So as per given information, we have the right equation as $x^2 - 10x + 9 = 0$.

Sol. 54(c)

$$(x+p)(x+q) - k = 0$$

$$\Rightarrow x^2 + qx + px + pq - k = 0$$

$$\Rightarrow x^2 + (p+q)x + pq - k = 0 \dots(i)$$

For m and n to be roots the equation should be

$$x^2 - (m+n)x + m.n = 0 \quad \dots(ii)$$

On comparing Eqs. (i) and (ii), we get

$$p+q = -(m+n) \quad \dots(iii)$$

$$\text{and } pq - k = m.n \quad \dots(iv)$$

$$\text{Also, } (x-m)(x-n) + k = 0$$

$$\Rightarrow x^2 - nx - mx + mn + k = 0$$

$$\Rightarrow x^2 - (m+n)x + mn + k = 0$$

$$\Rightarrow x^2 + (p+q)x + pq - k + k = 0$$

(using Eqs. (iii) (iv))

$$\Rightarrow x^2 + (p+q)x + pq = 0$$

$$\Rightarrow x^2 - [(-p) - (-q)] + [(-p)(-q)] = 0$$

[using Eqs. (iii) and (iv)]

$$\Rightarrow x^2 + (p+q)x + pq = 0$$

$$\Rightarrow x^2 - [(-p) - (-q)] + [(-p)(-q)] = 0$$

Hence, $-p$ and $-q$ are the required roots.

Sol. 55(b)

We have, $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$

$$(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$$

$$\Rightarrow \left(\frac{x^2+2}{x}\right)^2 - 6\left(\frac{x^2+2}{x}\right) + 8 = 0$$

$$\Rightarrow \left(\frac{x^2+2}{x} - 2\right)\left(\frac{x^2+2}{x} - 4\right) = 0$$

$$\Rightarrow \frac{x^2+2}{x} = 2 \Rightarrow x^2 + 2 = 2x$$

$$\Rightarrow x^2 - 2x + 2 = 0 \Rightarrow D \leq 0$$

$$\Rightarrow \frac{x^2+2}{x} = 4 \Rightarrow x^2 + 2 = 4x$$

$$\Rightarrow x^2 - 4x + 2 = 0 \Rightarrow D \geq 0$$

sum of roots of first equation is 2 and second is 4,

sum of all roots = 6.

Sol. 56(c)

$$\text{Equation: } ax^2 + bx + c = 0$$

$$\alpha + \beta = a^2 + \beta^2 \quad (\text{given})$$

Sum of roots, $(\alpha + \beta) = -b/a$

Product of roots, $(\alpha\beta) = c/a$

$$(\alpha+\beta) = (\alpha+\beta)^2 - 2\alpha\beta$$

$$\{ \because \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \}$$

$$\text{or } -\frac{b}{a} = \frac{b^2}{a^2} - \frac{2c}{a}$$

$$\text{or } -ab = b^2 - 2ac$$

$$\text{or } b^2 + ab = 2ac$$

Sol. 57(a)

$$x^2 - nx + m = 0$$

$$\alpha - \beta = 1 \dots (i)$$

$$\text{Sum of roots } (\alpha + \beta) = n$$

$$\text{Product of roots } (\alpha\beta) = m$$

Squaring of both sides in Eq. (i)

$$(\alpha - \beta)^2 = (1)^2$$

$$\text{or } \alpha^2 + \beta^2 - 2\alpha\beta = 1$$

$$\text{or } (\alpha + \beta)^2 - 2\alpha\beta - 2\alpha\beta = 1$$

$$\{ \because (\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta \}$$

$$\text{or } n^2 - 4m - 1 = 0.$$

Sol. 58(a)

$$\text{Equation: } x^2 - 3|x| + 2 = 0$$

$$x > 0 \Rightarrow +(\text{Taking positive value})$$

$$x^2 - 3x + 2 = 0$$

$$x^2 - 2x - x + 2 = 0$$

$$x = 1, 2$$

$$x < 0 \Rightarrow \text{Taking negative value}$$

$$x^2 + 3x + 2 = 0$$

$$x^2 + 2x + x + 2 = 0$$

$$x = -1, -2$$

Hence, number of real roots are 4.

Sol. 59(c)

$$\alpha + \beta = -b = -ve \quad \{ \because b > 0 \} \quad \dots (i)$$

$$\text{and } \alpha\beta = c = -ve \quad \{ \because c > 0 \} \quad \dots (ii)$$

From (ii), **Case I.** $\alpha > 0$ and $\beta < 0 \Rightarrow \alpha\beta = -ve$

Now, in (i), if $\alpha > 0$, then $\beta < 0$

$\Rightarrow \alpha > \beta$ but it is given that $\alpha < \beta$

$$\therefore \alpha \neq 0$$

Case II. If $\alpha < 0$, then $\beta > 0 \Rightarrow \alpha\beta = -ve$

Now, in (i), if $\alpha < 0$, $\beta > 0$ and $\alpha < \beta$

Which satisfies the given condition

Thus, $\alpha < 0 \Rightarrow -\alpha > 0$

If $\alpha < \beta$ then $-\alpha > \beta$ (Use number line)

$\therefore$ Statement 1 is correct.

From above, $\alpha < \beta$ and $-\alpha > \beta$

$$\Rightarrow \alpha < \beta < -\alpha$$

$$\Rightarrow \beta < |\alpha|$$

$\therefore$ statement 2 is correct.

Sol. 60(b)

$$\alpha + \beta = -(b)$$

Since $b > 0$, then $\alpha + \beta = -(+ve)$

$$\alpha + \beta = -ve$$

$$\text{and } \alpha\beta = c$$

Since $b > 0$, then $\alpha\beta = -ve$

Case I. $\alpha + \beta + \alpha\beta = (-ve) + (-ve) = -ve$

$$\Rightarrow \alpha + \beta + \alpha\beta < 0$$

$\therefore$ Statement 1 is not correct.

Case II. $\alpha^2\beta + \beta^2\alpha = \alpha\beta(\alpha + \beta)$

$$= (-ve)(-ve) = +ve$$

$$\alpha^2\beta + \beta^2\alpha > 0$$

$\therefore$ Statement 2 is correct.

Sol. 61(b)

Let the roots of the equation are 2α and α

$$\text{Product} = 2\alpha \cdot \alpha = \frac{1}{l-m}$$

$$\alpha^2 = \frac{1}{2(l-m)}$$

$$\text{and sum} = 3\alpha = \frac{-l}{(l-m)} \Rightarrow \alpha = -\frac{l}{3(l-m)}$$

$$\frac{2l^2}{9(l-m)^2} = \frac{1}{l-m} \Rightarrow l = \frac{9 \pm \sqrt{81-72m}}{4}$$

l is real

$$D \geq 0$$

$$81-72m \geq 0 \Rightarrow m \leq 9/8$$

Sol. 62(d)

$$x^2 + bx + c = 0$$

Roots of the equation are $\tan\alpha$ and $\tan\beta$

Then,

$$\tan\alpha + \tan\beta = -b$$

$$\tan\alpha \cdot \tan\beta = c$$

Now,

$$\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha \cdot \tan\beta}$$

$$\text{or } \tan(\alpha + \beta) = \frac{-b}{1-c} \Rightarrow -b(1-c)^{-1} = b(c-1)^{-1}$$

Sol. 63(b)

$$\sin(\alpha + \beta) \cdot \sec\alpha \sec\beta = \frac{\sin\alpha \cos\beta + \cos\alpha \sin\beta}{\cos\alpha \cos\beta}$$

$$= \tan\alpha + \tan\beta$$

$$\sin(\alpha + \beta) \cos\alpha \sec\beta = -b$$

Sol. 64(a)

$$\text{If } x^2 - px + 4 > 0$$

Then for all real values of x

$$b^2 - 4ac < 0$$

$$\text{or } (-p)^2 - 4(1)(4) < 0$$

$$\text{or } p^2 - 16 < 0$$

$$\text{or } |p| < 4$$

Sol. 65(d)

Roots of the equation,

$$x^2 - 2kx + k^2 - 4 = 0 \text{ are}$$

$$\frac{-(-2k) \pm \sqrt{(-2k)^2 - 4(k^2 - 4)}}{2 \times 1}$$

$$= \frac{2k \pm \sqrt{16}}{2} = k \pm 2$$

As given roots are lie between -3 and 5 , so

$$-3 < k + 2 < 5 \text{ and } -3 < k - 2 < 5$$

$$\Rightarrow -5 < k < 3 \text{ and } -1 < k < 7$$

$$\therefore -1 < k < 3$$

Sol. 66(d)

$$\text{We have } x^2 - (1-2a^2)x + (1-2a^2) = 0$$

For real roots, $D \geq 0$

$$\therefore (1-2a^2)^2 - 4(1-2a^2) \geq 0$$

$$\Rightarrow (1-2a^2)(1-2a^2-4) \leq 0$$

$$\Rightarrow (1-2a)(2a^2+3) \leq 0 \quad [\because 2a^2+3 > 0]$$

$$\Rightarrow (1-2a^2) \leq 0 \Rightarrow a^2 \geq 1/2$$

Sol. 67(a)

We have,

α, β as the roots of the equation,

$$x^2 - (1-2a^2)x + (1-2a^2) = 0$$

$$\therefore \alpha + \beta = 1-2a^2$$

$$\alpha\beta = 1-2a^2$$

$$\text{Now, } \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$$

$$\Rightarrow \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{(\alpha + \beta)^2}{(\alpha\beta)^2} - \frac{2}{\alpha\beta}$$

$$\Rightarrow \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{(1-2a^2)^2}{(1-2a^2)^2} - \frac{2}{1-2a^2}$$

$$\Rightarrow \frac{1}{\alpha^2} + \frac{1}{\beta^2} = 1 - \frac{2}{1-2a^2}$$

$$\text{Since, } \frac{1}{\alpha^2} + \frac{1}{\beta^2} < 1$$

$$\therefore 1 - \frac{2}{1-2a^2} < 1$$

$$\Rightarrow \frac{2}{2a^2-1} < 0$$

$$\Rightarrow 2a^2 < 1 \quad \{ \because 2 > 0 \}$$

$$\Rightarrow a^2 < 1/2$$

Sol. 68(c)

$\therefore -2$ and β are the roots of the equation

$$2x^2 + 3x - \alpha = 0$$

$$\therefore 2(-2)^2 + 3(-2) - \alpha = 0$$

$$\Rightarrow 8 - 6 - \alpha = 0 \Rightarrow \alpha = 2$$

Sol. 69(a)

Since, $\alpha = 2$

$\therefore$ Equation becomes

$$2x^2 + 3x - 2 = 0$$

$$\Rightarrow 2x^2 + 4x - x - 2 = 0$$

$$\Rightarrow (2x-1)(x+2) = 0$$

$$\Rightarrow x = -2, x = 1/2$$

$$\therefore \beta = 1/2$$

The roots of second equation are given by

$$x^2 - 3mx + 2m^2 = 0$$

$$\Rightarrow x^2 = 2mx - mx + 2m^2 = 0$$

$$\Rightarrow (x-2m)(x-m) = 0$$

$$\Rightarrow x = 2m, x = m$$

Since, both roots are positive

$$\therefore m > 0$$

$$\therefore \beta, 2, 2m \text{ are in P.}$$

$$\therefore 4 = 2m\beta \Rightarrow 4 = 2m \times 1/2 \Rightarrow m = 4$$

$$\therefore \beta \sqrt{m} = \frac{1}{2} \sqrt{4} = \frac{1}{2} \times 2 = 1$$

Sol. 70(a)

We have,

$$c > 0 \text{ and } 4a + c < 2b$$

$$\text{Let } f(x) = ax^2 - bx + c$$

$$\text{Then, } f(0) = c > 0$$

$$\Rightarrow f(2) = 4a - 2b + c$$

$$\Rightarrow f(2) < 0$$

$$\therefore f(0), f(2) < 0$$

Hence, one of the root is lie between $(0, 2)$

Sol. 71(c)

According to the question,

$$\alpha + \beta = -k$$

$$\alpha\beta = 1$$

$$\text{and } \alpha - \beta < \sqrt{5}$$

$$\text{or } (\alpha - \beta)^2 < 5 \quad \dots (i)$$

$$\text{Now, } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$\text{or } (\alpha - \beta)^2 = k^2 - 4 \quad \dots (ii)$$

From equations (i) and (ii),

$$k^2 - 4 < 5$$

$$\text{or } k^2 < 9 \Rightarrow -3 < k < 3$$

Also, it is given that $|k| \geq 2$

$\therefore$ The required interval in which k lies, is

$$(-3, -2] \cup [2, 3)$$

Sol. 72(a)

If α, β are the roots of $x^2 + px + q = 0$

and γ, δ are the roots of $x^2 + lx + m = 0$

then

$$\frac{(\alpha + \beta)}{(\gamma + \delta)} = \frac{\sqrt{\alpha\beta}}{\sqrt{\gamma\delta}} \quad (\text{Remember})$$

$$\text{or } \frac{(-p)^2}{(-l)^2} = \frac{q}{m} \text{ or } p^2 m = l^2 q.$$

Sol. 73(b)

Since, $\cot\alpha$ and $\cot\beta$ are the roots of the equation

$$x^2 + bx + c, \text{ therefore}$$

$$\cot\alpha + \cot\beta = -b$$

$$\text{and } \cot\alpha \cdot \cot\beta = c$$

Now, $\cot(\alpha + \beta) =$

$$\frac{\cot\alpha \cot\beta - 1}{\cot\alpha + \cot\beta} = \frac{c-1}{-b} = \frac{1-c}{b}$$

Sol. 74(c)

⇒ Both the roots are imaginary i.e. complex

Sol. 75(b)

$$(q-r)x^2 + (r-p)x + (p-q) = 0$$

$$\text{The product of the roots} = \frac{p-q}{q-r}$$

sum of coefficients is zero, so one root is

$$\frac{p-q}{q-r}$$

definitely 1. so other must be

Sol. 76(d)

$$|x-3|^2 + |x-3| - 2 = 0$$

$$\text{Case I. } (x-3)^2 + (x-3) - 2 = 0$$

$$\text{or } y^2 + y - 2 = 0 \quad \{\text{where } y = (x-3)\}$$

$$\text{or } (y+2)(y-1) = 0$$

$$\text{or } y = -2 \text{ or } y = 1$$

$$\text{or } (x-3) = -2 \text{ or } (x-3) = 1$$

$$\text{or } x = 1 \text{ or } x = 4$$

But $x=1$ does not satisfy the given equation, so $x = 4$.

$$\text{Case II. } [-(x-3)]^2 - (x-3) - 2 = 0$$

$$\text{or } (x-3)^2 - (x-3) - 2 = 0$$

$$\text{or } y^2 - y - 2 = 0 \quad \{\text{where } y = (x-3)\}$$

$$\text{or } (y-2)(y+1) = 0$$

$$\text{or } y = 2 \text{ or } y = -1$$

$$\text{or } x-3 = 2 \text{ or } x-3 = -1$$

$$\text{or } x = 5 \text{ or } x = 2$$

But $x = 5$ does not satisfy the given equation, so $x = 2$. Thus, the required sum $= 4 + 2 = 6$

Sol. 77(a)

α and β are the roots of $3x^2 + 2x + 1 = 0$

$$\alpha + \beta = -\frac{2}{3} = -\frac{1}{3}$$



$$\text{Now} =$$

$$= -8/3 = \text{sum}$$

$$\text{and} = \left(\alpha + \frac{1}{\beta}\right) \cdot \left(\beta + \frac{1}{\alpha}\right) = \alpha\beta + \frac{1}{\alpha\beta} + 2$$

$$= \frac{1}{3} + 3 + 2 = \frac{16}{3} = \text{product}$$

$$\therefore \text{required equation is: } x^2 - \left(-\frac{8}{3}\right)x + \frac{16}{3} = 0$$

$$\text{Or } 3x^2 + 8x + 16 = 0$$

Sol. 78(c)

$$\text{In } \Delta PQR, \angle R = \pi/2 \Rightarrow \angle P + \angle Q = \pi/2$$

$$\therefore \tan\left(\frac{P+Q}{2}\right) = \tan\left(\frac{\pi}{4}\right)$$

$$\text{or } \frac{\tan\left(\frac{P}{2}\right) + \tan\left(\frac{Q}{2}\right)}{1 - \tan\left(\frac{P}{2}\right)\tan\left(\frac{Q}{2}\right)} = 1 \text{ or } \frac{-\frac{b}{a}}{1 - \frac{c}{a}} = 1$$

$$\left\{ \because \tan\left(\frac{P}{2}\right) \text{ and } \tan\left(\frac{Q}{2}\right) \text{ are the roots of } ax^2 + bx + c = 0 \right\}$$

$$\text{or } \frac{-b}{a-c} = 1 \text{ or } a+b=c$$

Sol. 79(c)

for real root $b^2 - 4ac \geq 0$

$$\Rightarrow 16 + 4 \log_3 P \geq 0$$

$$\Rightarrow 4 \log_3 P \geq -16$$

$$\Rightarrow \log_3 P \geq -4$$

$$\Rightarrow P \geq 1/81$$

Sol. 80(a)

$$|1-x| + x^2 = 5$$

$$\Rightarrow 1-x+x^2=5, x < 1$$

$$\text{or } -1+x+x^2=5, x \geq 1$$

$$\Rightarrow x^2 - x - 4 = 0, x < 1 \text{ or } x^2 + x - 6 = 0, x \geq 1$$

$$\Rightarrow x = \frac{-1-\sqrt{17}}{2} \text{ or } x = 2$$

Equation has a rational root and an irrational root.

Sol. 81(b)

$$f(x) > 0$$

$$\text{let } f(x) = x^2 + 1$$

$$g(x) = f(x) + f'(x) + f''(x)$$

$$g(x) = x^2 + 1 + 2x + 2$$

$$g(x) = x^2 + 2x + 3 = (x+1)^2 + 2 > 0$$

$$g(x) > 0$$

Sol. 82(b)

$$\frac{d_1}{d} = (\text{ratio of coefficient of } x)^2 = \frac{b^2}{q^2}$$

Sol. 83(a)

$$\cos \alpha \cdot \cos \beta = -3/4$$

$$\Rightarrow \frac{1}{\cos \alpha \cdot \cos \beta} = \sec \alpha \cdot \sec \beta = -\frac{4}{3}$$

Sol. 84(d)

$$\alpha + \beta = -\alpha, \alpha\beta = -\beta \Rightarrow \alpha\beta + \beta = 0$$

$$\Rightarrow (\alpha+1)\beta = 0 \Rightarrow \alpha = -1 (\beta \neq 0)$$

$$\Rightarrow 2\alpha + \beta = 0$$

$$\Rightarrow \beta = 2$$

$$\therefore -x^2 + \alpha x + \beta = -x^2 - x + 2$$

$$\text{Greatest value} = -\frac{1+8}{-4} = \frac{9}{4}$$

Sol. 85(b)

By definition of rational function

Option (b) is correct

Sol. 86(d)

$$|x^2 - x - 6| = x + 2$$

$$|(x-3)(x+2)| = x + 2$$

$$\text{for } x \leq -2 \text{ or } x \geq 3$$

$$\Rightarrow x^2 - x - 6 = x + 2$$

$$\Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow x = 4, -2$$

$$\text{for } -2 < x < 3$$

$$\Rightarrow -(x^2 - x - 6) = x + 2$$

$$\Rightarrow x^2 - 4 = 0$$

$$\Rightarrow x = 2, -2$$

$$\text{so } x = 2, -2, 4$$

Sol. 87(b)

If all coefficients are positive then both roots will be negative.

Sol. 88(a)

$$x^2 + px + q = 0$$

$$\tan 45^\circ = \tan (19^\circ + 26^\circ) = \frac{\tan 19^\circ + \tan 26^\circ}{1 - \tan 19^\circ \tan 26^\circ}$$

$$\Rightarrow 1 = \frac{-p}{1-q} \Rightarrow 1-q = -p \Rightarrow q-p=1$$

$$\text{are } \tan 19^\circ \text{ and } \tan 26^\circ$$

Sol. 89(a)

$$x^2 + 9|x| + 20 = 0$$

$$(|x|+4)(|x|+5) = 0$$

$$|x| = -4 \text{ or } -5 \text{ (not possible)}$$

number of real roots is zero.

Sol. 90(c)

$$x \leq 4 \text{ and } x \leq -4$$

$$\text{common interval } x \leq -4$$

$$y \geq 0 \text{ and } y \leq 0$$

$$\text{common interval } y = 0$$

Sol. 91(a)

$$x^2 + 3|x| + 2 = 0$$

$$(|x|+2)(|x|+1) = 0$$

$$|x| = -2 \text{ or } -1 \text{ (not possible)}$$

number of real roots is zero.

Sol. 92(c)

$$x^2 - (p+q)x + pq - r^2/4 = 0$$

$$\text{for real root } b^2 - 4ac \geq 0$$

$$(p+q)^2 - 4(pq - \frac{r^2}{4}) \geq 0$$

$$(p-q)^2 + \frac{r^2}{4} \geq 0$$

it is always correct.

for equal root $p = q$ and $r = 0$

Sol. 93(c)

for real root $b^2 - 4ac \geq 0$

$$q^2 - 4p \geq 0$$

$$q^2 \geq 4p$$

$$\text{if } p=1, \text{ then possible value of } q = \{2, 3, 4\}$$

$$\text{if } p=2, \text{ then possible value of } q = \{3, 4\}$$

$$\text{if } p=3, \text{ then possible value of } q = \{4\}$$

$$\text{if } p=4, \text{ then possible value of } q = \{4\}$$

total 7 solution.

Sol. 94(c)

sum of coefficients is zero so its one root

$$\text{is } 1 \text{ and another is } \frac{c(a-b)}{a(b-c)}$$

$$\text{both are equal } \frac{c(a-b)}{a(b-c)} = 1$$

$$2ac = ab + bc$$

this condition is of HP.

Sol. 95(c)

it is only possible when

$$x^2 - 3x + 2 < 0$$

$$(x-1)(x-2) < 0$$

$$1 < x < 2$$

Sol. 96(c)

for real root $b^2 - 4ac \geq 0$

$$m^2 - 8 \geq 0$$

$$m^2 \geq 8$$

Sol. 97(b)

$$x = 2 + \frac{1}{x}$$

$$x^2 - 2x - 1 = 0$$

$$x = \sqrt{2} + 1$$

Sol. 98(b)

$$(x-a)(x-b) > 0$$

$$\text{i.e. } x < a \text{ or } x > b$$

Sol. 99(b)

$$p+q = 30 \text{ and } pq = 221$$

$$(p+q)^3 = p^3 + q^3 + 3pq(p+q)$$

$$(30)^3 = p^3 + q^3 + 3(221)(30)$$

$$p^3 + q^3 = 7110$$

Sol. 100(d)

$$\alpha\beta = \alpha^2\beta^2$$

$$\alpha\beta(\alpha\beta-1) = 0$$

$$\alpha\beta = 0 \text{ or } 1$$

$$\text{Now } \alpha + \beta = \alpha^2 + \beta^2$$

$$\alpha + \beta = (\alpha + \beta)^2 - 2\alpha\beta$$

put $\alpha\beta = 0$ then there are two possible values of $\alpha + \beta$

$$\alpha + \beta = (\alpha + \beta)^2$$

$$\alpha + \beta = 0 \text{ or } 1$$

put $\alpha\beta = 1$ then there are also two possible values of $\alpha + \beta$

$$\alpha + \beta = (\alpha + \beta)^2 - 2$$

$$\alpha + \beta = 2 \text{ or } -1$$

so 4 possible quadratic equations are there.

Sol. 101(d)

given $1.5 \leq x$

$$3 \leq 2x \text{ or } 2x \geq 3 \Rightarrow (2x-3) \geq 0 \dots\dots(i)$$

$$\text{and } x \leq 4.5$$

$$2x \leq 9 \Rightarrow (2x-9) \leq 0 \dots\dots(ii)$$

multiply (i) and (ii)

$$(2x-3)(2x-9) \leq 0$$

Sol. 102(a)

If α and β are the roots of the equation

$$4x^2 + 2x - 1 = 0,$$

$$\alpha + \beta = -1/2$$

$$\beta^2 + 2\beta - 1 = 0,$$

$$4\left(-\frac{1}{2} - \alpha\right)^2 + 2\beta - 1 = 0$$

by solving it

$$\beta = -2\alpha^2 - 2\alpha$$

Sol. 103(d)

Roots of the equation $x^2 + 2x + k = 0$ are real then $D \geq 0$

$$\Rightarrow 4 - 4k \geq 0$$

$$\Rightarrow k \leq 1$$

Sol. 104(c)

If one root of equation is reciprocal to other then $a = c$

$$\text{So } k = 5$$

Sol. 105(c)

$$\alpha - \beta = 1$$

$$\sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = 1$$

$$\left(\frac{5k+1}{4}\right)^2 - 4\left(\frac{5k}{4}\right) = 1$$

$$\frac{25k^2 + 10k + 1}{16} = 5k + 1$$

$$25k^2 - 70k - 15 = 0$$

$$K = 3, k = -1/5$$

Sol. 106(d)

Roots are real

$$b^2 - 4ac > 0$$

$$(k^2 + 5k)^2 - 4(3)(3k^2 - 5k) > 0$$

$$k^4 + 10k^3 + 25k^2 - 36k^2 + 60k > 0$$

$$k^4 + 10k^3 - 11k^2 + 60k > 0$$

$$k(k^3 + 10k^2 - 11k + 60) > 0$$

Both values of $k = 0, -5$

Do not satisfy given Inequality

So no such value of k exist

Roots are equal magnitude

But opposite sign so,

$$\alpha + \beta = 0$$

$$k^2 + 5k = 0$$

$$k = 0, k = -5$$

Sol. 107(b)

$$x^2 + px + q = 0$$

$$p + q = -p/1$$

$$\Rightarrow 2p = q \text{ is sum of roots} = -b/a$$

$$pq = (q)/1$$

$$\Rightarrow (p-1) = 0 \text{ product of roots} = c/a$$

$$\Rightarrow q=0 \text{ or } p=1$$

As p and q are non zero roots so $q \neq 0$

So we are left with $p = 1$ when $p = 1$

$$\Rightarrow 9 = -2$$

So ans is one (b) option

Sol. 108(c)

$$x^4 - 10x + 9 = 0$$

$$(x^2 - 9)(x^2 - 1) = 0$$

All values of x are 3, -3, 1, -1

Sum of absolute values = $3+3+1+1=8$

Sol. 109(d)

$$5x - 4y + 12 < 0$$

$$x + y < 2$$

$$x < 0, y > 0$$

By checking option.

Only $(-1, 2)$ satisfies these inequalities.

Sol. 110(d)

$$\alpha + \beta = -p \dots\dots\dots(1)$$

$$\alpha\beta = q \dots\dots\dots(2)$$

$$\alpha^3 + \beta^3 = -m \dots\dots\dots(3)$$

$$\alpha^3\beta^3 = n \dots\dots\dots(4)$$

By (3) and (4)

$$m+n = \alpha^3\beta^3 - (\alpha^3 + \beta^3)$$

$$= \alpha^3\beta^3 - [(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)]$$

$$= q^3 - [-p^3 - 3q(-p)]$$

$$= p^3 + q^3 - 3pq$$

Sol. 111(a)

$$\alpha + \beta = a + b$$

$$\alpha\beta = ab - c$$

$$\Rightarrow a + b = \alpha + \beta$$

$$ab = \alpha\beta + c$$

Equation will be

$$x^2 - \alpha x - \beta x + \alpha\beta + c = 0$$

Sol. 112(c)

Both roots are equal so $B^2 - 4AC = 0$

$$(a + b + c)^2 - 4(bc + ca) = 0$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca - 4bc - 4ca = 0$$

$$(a + b - c)^2 = 0$$

$$\Rightarrow a + b - c = 0$$

Sol. 113(d)

$$\alpha + \beta = 8 \dots(1)$$

$$\alpha\beta = q \dots(2)$$

$$\alpha^2 - \beta^2 = 16$$

$$(\alpha + \beta)(\alpha - \beta) = 16$$

$$\text{of } (\alpha - \beta) = 16$$

$$\alpha - \beta = 2 \dots(3)$$

By (1) & (2)

$$\alpha = 5, \beta = 3$$

$$\therefore \alpha\beta = q$$

$$\therefore q = 5 \times 3 = 15$$

Sol. 114(d)

$$x^2 - (\text{sum})x + (\text{product}) = 0$$

$$x^2 - 2x + 2 = 0.$$

$$x^2 - 3x + 3 = 0$$

$$x^2 - 4x + 4 = 0$$

infinite equations of this type are possible

Sol. 115(a)

Let p, q ($p > q$) be the roots of the quadratic equation $x^2 + bx + c = 0$ where $c > 0$.

$$\text{If } p^2 + q^2 - 11pq = 0,$$

$$(p-q)^2 = p^2 + q^2 - 2pq$$

$$(p-q)^2 = 11pq - 2pq = 9pq = 9c$$

$$p - q = 3\sqrt{c}$$

Sol. 116(b)

How many real numbers satisfy the equation

$$|x - 4| + |x - 7| = 15$$

$$\text{If } x > 7$$

$$x - 4 + x - 7 = 15$$

$$x = 13$$

$$\text{if } 4 < x < 7$$

$$x - 4 - x + 7 = 15$$

$$3 = 15 \text{ not possible}$$

$$\text{If } x < 4$$

$$-x + 4 - x + 7 = 15$$

$$x = -2$$

so two values of x are possible 13 and -2

Sol. 117(c)

$$1. \alpha + \beta = 0 \Rightarrow \alpha = -\beta$$

$$\alpha^2 + \beta^2 = 2 \Rightarrow \beta^2 + \beta^2 = 2$$

$$\beta^2 = 1$$

$$\beta = 1 \text{ then } \alpha = -1$$

$$\beta = -1 \text{ then } \alpha = 1$$

$$2. \alpha\beta^2 = -1, a = 0$$

$$a = 0 \text{ then } \alpha + \beta = 0 \Rightarrow \alpha = -\beta$$

$$\alpha^3 = -1$$

$$\alpha = -1$$

Both statements are individually sufficient to answer the question

Sol. 118(d)

$$\text{If } \alpha = 2 - i\sqrt{3} \text{ then } \beta = 2 - i\sqrt{3}$$

Then equation will be

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 - 4x + 7 = 0$$

$$a = -4 \text{ and } b = 7$$

$$a + b = -4 + 7 = 3$$

Sol. 119(b)

roots are real

$$b^2 - 4ac \geq 0$$

$$16 - 4k \geq 0$$

$$4 \geq k \dots\dots\dots(i)$$

both roots lie between 0 and 5

$$\text{so } f(0)f(5) > 0$$

$$k(5+k) > 0$$

$$\text{i.e. } k \in (-5, 0)$$

so k may be 1, 2, 3, 4

Sol. 120(b)

$$\Rightarrow (1-x)^4 + (5-x)^4 = 82$$

$$\Rightarrow (x-1)^4 + (x-5)^4 = 82$$

$$\text{let } x - 3 = y$$

$$\Rightarrow (y+2)^4 + (y-2)^4 = 82$$

$$\Rightarrow (y^4 + 4y^3 \cdot 2 + 6y^2 \cdot 2^2 + 4y \cdot 2^3 + 2^4) + (y^4 - 4y^3 \cdot 2 + 6y^2 \cdot 2^2 - 4y \cdot 2^3 + 2^4) = 82$$

$$\Rightarrow 2y^4 + 48y^2 + 32 = 82$$

$$\Rightarrow y^4 + 24y^2 - 25 = 0$$

$$\Rightarrow (y^2 + 25)(y^2 - 1) = 0$$

$$\text{so } y = 1, -1, 5i, -5i$$

$$\text{then } x = 4, 2, 3 + 5i, 3 - 5i$$

there is two real roots

Sol. 121(b)

from above solution

$$\text{sum of all roots} = 4 + 2 + 3 + 5i + 3 - 5i = 12$$

Sol. 122(b)

for nature of roots

$$B^2 - 4AC$$

$$(a + b + c)^2 - 4(a + b)c$$

$$\text{put } k = c/2$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca - 2(a + b)c$$

$$a^2 + b^2 + c^2 + 2ab$$

$$(a + b)^2 + c^2$$

$$\text{given that } c \neq 0$$

$$\text{so } (a + b)^2 + c^2 > 0$$

$$B^2 - 4AC > 0 \text{ so roots will be real and different.}$$

Sol. 123(c)

if sum of coefficients is 0 then one root will be

definitely 1 so let $\alpha = 1$

$$\text{and we know that } \alpha\beta = \frac{C}{A} \Rightarrow \beta = \frac{C}{a+b}$$

Sol. 124(c)

$$\text{if one root is } -\sqrt{2} \text{ then other will be } \sqrt{2}$$

$$\text{and if one root is } \sqrt{3} \text{ then other will be } -\sqrt{3}$$

$$\text{so all four roots are } -\sqrt{2}, \sqrt{2}, \sqrt{3}, -\sqrt{3}$$

then equation will be

$$(x - \sqrt{2})(x + \sqrt{2})(x - \sqrt{3})(x + \sqrt{3}) = 0$$

$$(x^2 - 2)(x^2 - 3) = 0$$

$$x^4 - 5x^2 + 6 = 0$$

now compare with given equation in question

$$a_0 = 6, a_1 = 0, a_2 = -5, a_3 = 0$$

Sol. 125(d)

roots are real

$$b^2 - 4ac \geq 0$$

$$\cos^2\beta - 4(\cos\beta - 1)\sin\beta \geq 0$$

$$\cos^2\beta + 4(1 - \cos\beta)\sin\beta \geq 0$$

we know that $\cos^2\beta$ and $\sin\beta$ is always positive for given interval and $(1 - \cos\beta)$ should be ≥ 0

Sol. 126(d)

a, b, c are in GP so $b^2 = ac$

we know that

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{-3b^2}}{2a} = \left(\frac{b}{a}\right) \frac{-1 \pm \sqrt{3}i}{2}$$

equation has imaginary roots $\left(\frac{b}{a}\right)\omega$ and $\left(\frac{b}{a}\right)\omega^2$

ratio of both above roots is $1 : \omega$

product of roots is $\left(\frac{b}{a}\right)^2 \omega^3 = \left(\frac{b}{a}\right)^2$

Sol. 127(c)

$x^2 + mx + n = Z$ for all x

put $x = 0$ then $n = Z$ so n must be integer.

put $x = 1$ then $1 + m + n = Z$ so m must be integer.

both statements are correct.

Sol. 128(c)

if sum of coefficients is 0 then one root will be definitely 1 so let $\alpha = 1$

both roots are equal so $\alpha = \beta = 1$

and we know that

$$\alpha\beta = \frac{C}{A} \Rightarrow \beta = \frac{c^2(a^2 - b^2)}{a^2(b^2 - c^2)} = 1$$

$$c^2(a^2 - b^2) = a^2(b^2 - c^2)$$

$$2a^2c^2 = a^2b^2 + b^2c^2$$

$$\frac{1}{b^2} = \frac{1}{a^2} + \frac{1}{c^2}$$

so a^2, b^2, c^2 are in HP.

Sol. 129(c)

$$\alpha + \beta = -\frac{B}{A}$$

$$\alpha + \alpha = -\frac{b^2(c^2 - a^2)}{a^2(b^2 - c^2)}$$

$$2\alpha = \frac{b^2(c^2 - a^2)}{a^2(c^2 - b^2)}$$

$$\alpha = \frac{b^2(c^2 - a^2)}{2a^2(c^2 - b^2)}$$

Sol. 130(a)

$$(x-1)^2 + (x-3)^2 + (x-5)^2 = 0$$

$$3x^2 - 18x + 35 = 0$$

$$b^2 - 4ac < 0$$

so both roots are imaginary.

Sol. 131(c)

n is a root of the equation $x^2 + px + m = 0$

$$\text{so } n^2 + pn + m = 0 \quad \dots\dots(i)$$

m is a root of the equation $x^2 + px + n = 0$

$$\text{so } m^2 + pm + n = 0 \quad \dots\dots(ii)$$

subtract above equations

$$n^2 - m^2 + p(n - m) + (m - n) = 0$$

$$(n - m)(n + m + p - 1) = 0$$

$$\text{so } m + n + p = 1$$

Sol. 132(b)

$$g(x) = 9x - 8\sqrt{x} - 1$$

$$g(x) = 9x - 9\sqrt{x} + \sqrt{x} - 1$$

$$g(x) = 9\sqrt{x}(\sqrt{x} - 1) + 1(\sqrt{x} - 1)$$

$$g(x) = (\sqrt{x} - 1)(9\sqrt{x} + 1) = 0$$

so $x = 1$ is only one real root.

Sol. 133(b)

$$\text{Given that } \alpha - \beta = 2\sqrt{3}$$

$$\text{From equation } x^2 - kx + k = 0$$

$$\alpha + \beta = k \text{ and } \alpha\beta = k$$

$$\text{We know that } \alpha - \beta = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$$

$$2\sqrt{3} = \sqrt{k^2 - 4k}$$

$$12 = k^2 - 4k$$

$$k^2 - 4k - 12 = 0,$$

$$k = 6, -2$$

Sol. 134 (a)

$$x + \frac{5}{y} = 4 \quad \dots\dots (i)$$

$$y + \frac{5}{x} = -4 \quad \dots\dots (ii)$$

Add both equations

$$x + y + \frac{5}{x} + \frac{5}{y} = 0$$

$$(x+y) + 5\left(\frac{1}{x} + \frac{1}{y}\right) = 0$$

$$(x+y)\left(1 + \frac{1}{xy}\right) = 0$$

$$x + y = 0$$

Sol. 135 (b)

$$\text{let } k = \log_{0.5}(a^2) \text{ so } \frac{1}{k} = \log_{a^2}(0.5)$$

$$x^2 + kx + k^4 = 0$$

$$\alpha + \beta = -k$$

$$\alpha \cdot \beta = k^4$$

$$\beta^2 = \frac{\alpha}{k} \quad \dots\dots(iii)$$

Multiply both side with β

$$\beta^3 = \frac{\alpha\beta}{k} = \frac{k^4}{k} = k^3$$

$$\beta = k = \log_{0.5}(a^2)$$

Sol. 136 (c)

Put $\beta = k$ in equation (i)

$$\alpha + k = -k$$

$$\alpha = -2k$$

$$\alpha = -2\beta$$

Sol. 137 (d)

$$abx^2 + bcx + ca = cax^2 + abx + bc$$

$$a(b-c)x^2 + b(c-a)x + c(a-b) = 0$$

sum of coefficients is zero so its one root

$$\text{is 1 and another is } \frac{c(a-b)}{a(b-c)}$$

$$\text{both are equal } \frac{c(a-b)}{a(b-c)} = 1$$

$$2ac = ab + bc$$

$$\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$

this condition is of HP.

Sol. 138 (c)

By above solution

$$\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$

this condition is of HP.

You can access the solutions of PYQ through 6 Youtube video:

1. [Equation 2011 to 2012 | NDA mathematics previous year questions by Sandeep Brar](#)

2. [Equation 2013 to 2014 | NDA mathematics previous year questions by Sandeep Brar](#)

3. [Equation 2015 to 2017 | NDA mathematics previous year questions by Sandeep Brar](#)

4. [Equation 2018 to 2019 | NDA mathematics previous year questions by Sandeep Brar](#)

5. [Equation 2020 to 2022 | NDA mathematics previous year questions by Sandeep Brar](#)

6. [Equation 2023 to 2025 | NDA mathematics previous year questions by Sandeep Brar](#)